

ICPC TRD

Main (1), Graphs (1), Trees (6), Range Queries (10), Strings (14), Dynamic Programming (17), Mathematics (19), Geometry (21), Bit Operations (24), Extra (24),

Main

```
// #pragma GCC optimize("O3,unroll-loops")
// #pragma GCC target("avx2,bmi,bmi2,lzcnt,popcnt")

#include <bits/stdc++.h>
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace std;
using namespace __gnu_pbds;

#define fastio ios::sync_with_stdio(false); cin.tie(0);
cout.tie(0)
#define all(v) v.begin(), v.end()
#define rall(v) v.rbegin(), v.rend()
#define pb push_back
#define fi first
#define se second

typedef long long ll;
typedef long double ld;
typedef pair<int,int> pii;
typedef vector<int> vi;
typedef vector<ll> vll;

const int INF = 1e9;
const ll LINF = 1e18;
const int MOD = 1e9+7;
const double EPS = 1e-9;

void solve(){}

int main() {
    #ifdef ICESI
        freopen("input.txt", "r", stdin);
    #endif
    fastio;
    int t = 1;
    // cin >> t;
    while (t--) {
        solve();
    }
    return 0;
}
```

Graphs

```
/**
 * Uso[1]: dx4, dy4, dx8, dy8, isValid(x, y, R, C);
 * Direcciones de movimiento en grillas 4/8-conexas y
 * validacion de coordenadas dentro de la grilla.
 * Complejidad: O(1).
 */
const int dx4[4] = {1, -1, 0, 0};
const int dy4[4] = {0, 0, 1, -1};
const int dx8[8] = {1, 1, 0, -1, -1, -1, 0, 1};
const int dy8[8] = {0, 1, 1, 1, 0, -1, -1, -1};
```

```
#define isValid(x, y, R, C) ((x) >= 0 && (x) < (R) && (y) >= 0 && (y) < (C))

/**
 * Uso: Edge<T> e = {from, to, id, weight};
 * Arista con extremos, identificador opcional y peso
 * generico; usada internamente por Graph.
 * Complejidad: O(1) por arista.
 */
template <typename T = ll> struct Edge { int from, to,
    id; T weight; };

/**
 * Uso[0]: DSU dsu(n); dsu.find(x); dsu.unite(a, b); dsu
    .component_size(x); dsu.comps; dsu.get_components()
    ;
 * Union-find con union por tamaño y acceso a los
    miembros de cada componente.
 * Complejidad: find y unite casi O(α(N)) amortizado.
 */
struct DSU {
    vector<int> p, sz;
    vector<vector<int>> members;
    int comps;
    DSU(int n) : p(n), sz(n, 1), members(n), comps(n) {
        iota(all(p), 0);
        for (int i = 0; i < n; i++) members[i].pb(i);
    }
    int find(int x) { return p[x] == x ? x : p[x] = find(p[x]); }
    bool unite(int a, int b) {
        a = find(a); b = find(b);
        if (a == b) return false;
        if (sz[a] < sz[b]) swap(a, b);
        p[b] = a; sz[a] += sz[b];
        members[a].insert(members[a].end(), all(members[b]))
        ;
        vector<int>().swap(members[b]);
        comps--;
        return true;
    }
    const vector<int>& component_nodes(int x) { return
        members[find(x)]; }
    int component_size(int x) { return sz[find(x)]; }
    vector<vector<int>> get_components(int first = 0)
    {
        const {
            vector<vector<int>> res;
            for (int i = first; i < p.size(); i++) if (p[i] == i
                && !members[i].empty()) res.pb(members[i]);
            return res;
        }
    };

/**
 * Uso[1]: Graph<ll> g(n); g.add_undirected_edge(u, v, w
    ); g.add_directed_edge(u, v, w, id);
 * Grafo con lista de adyacencia de índices a aristas y
    pesos genericos.
```

```
* Complejidad: agregar arista O(1).
*/
template <typename T = ll> struct Graph {
    int n;
    vector<vector<int>> adj; // indices a edges
    vector<Edge<T>> edges;
    Graph(int n) : n(n) { adj.resize(n + 1); }
    void add_directed_edge(int u, int v, T w = 1, int id =
        -1) {
        adj[u].pb(edges.size());
        edges.pb({u, v, id, w});
    }
    void add_undirected_edge(int u, int v, T w = 1, int id
        = -1) {
        add_directed_edge(u, v, w, id);
        add_directed_edge(v, u, w, id);
    }
};

/**
 * Uso[1]: auto dist = Dijkstra<ll>().run(g, src); auto
    [dist, parent] = run_with_parents(g, src);
 * Caminos minimos desde un origen con pesos no
    negativos; opcionalmente reconstruye el camino.
 * Complejidad: O(E log V).
 */
template <typename T = ll> struct Dijkstra {
    // O(E Log V)
    vector<T> run(const Graph<T>& g, int src) {
        vector<T> dist(g.n + 1, LINF);
        priority_queue<pair<T, int>, vector<pair<T, int>>,
            greater<>> pq;
        dist[src] = 0; pq.push({0, src});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int id : g.adj[u]) {
                auto& e = g.edges[id];
                if (dist[u] + e.weight < dist[e.to]) {
                    dist[e.to] = dist[u] + e.weight;
                    pq.push({dist[e.to], e.to});
                }
            }
        }
        return dist;
    }
    // Retorna {distancias, padres} para reconstruir el
    camino
    pair<vector<T>, vi> run_with_parents(const Graph<T>& g,
        int src) {
        vector<T> dist(g.n + 1, LINF);
        vi p(g.n + 1, -1);
        priority_queue<pair<T, int>, vector<pair<T, int>>,
            greater<>> pq;
        dist[src] = 0; pq.push({0, src});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
```

```

    for (int id : g.adj[u]) {
        auto& e = g.edges[id];
        if (dist[u] + e.weight < dist[e.to]) {
            dist[e.to] = dist[u] + e.weight;
            p[e.to] = u;
            pq.push({dist[e.to], e.to});
        }
    }
    return {dist, p};
}
static vi restore_path(int target, const vi& p) {
    vi path;
    for (int v = target; v != -1; v = p[v]) path.pb(v);
    reverse(all(path));
    return path;
}
};

/**
 * Uso[1]: auto dist = ZeroOneBfs<LL>().run(g, src);
 * Caminos minimos con pesos restringidos a {0,1}
 * mediante un deque.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct ZeroOneBfs {
    vector<T> run(const Graph<T>& g, int src) {
        vector<T> dist(g.n + 1, LINF);
        deque<int> dq;
        dist[src] = 0; dq.pb(src);
        while (!dq.empty()) {
            int u = dq.front(); dq.pop_front();
            for (int id : g.adj[u]) {
                auto& e = g.edges[id];
                if (dist[u] + e.weight < dist[e.to]) {
                    dist[e.to] = dist[u] + e.weight;
                    e.weight ? dq.pb(e.to) : dq.push_front(e.to);
                }
            }
        }
        return dist;
    }
};

/**
 * Uso[1]: auto [es_bipartito, colores] = Bipartite<LL>().check(g);
 * Verifica si el grafo es bipartito y colorea cada
 * componente con 0/1.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct Bipartite {
    pair<bool, vi> check(const Graph<T>& g) {
        vi color(g.n + 1, -1);
        bool ok = true;
        for (int i = 1; i <= g.n && ok; i++) if (color[i] == -1) {
            queue<int> q;
            q.push(i); color[i] = 0;
            while (!q.empty()) {
                int u = q.front(); q.pop();
                for (int id : g.adj[u]) {
                    int v = g.edges[id].to;

```

```

                    if (color[v] == -1) color[v] = color[u] ^ 1, q
                    .push(v);
                    else if (color[v] == color[u]) ok = false;
                }
            }
        }
        return {ok, color};
    }
};

/**
 * Uso[1]: auto dist = FloydWarshall<LL>().run(g);
 * Caminos minimos entre todos los pares de nodos.
 * Complejidad:  $O(V^3)$  tiempo,  $O(V^2)$  espacio.
 */
template <typename T = ll> struct FloydWarshall {
    vector<vector<T>> run(const Graph<T>& g) {
        vector<vector<T>> dist(g.n + 1, vector<T>(g.n + 1, LINF));
        for (int i = 1; i <= g.n; i++) dist[i][i] = 0;
        for (auto& e : g.edges) dist[e.from][e.to] = min(
            dist[e.from][e.to], e.weight);
        for (int k = 1; k <= g.n; k++)
            for (int i = 1; i <= g.n; i++)
                for (int j = 1; j <= g.n; j++)
                    if (dist[i][k] < LINF && dist[k][j] < LINF)
                        dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);
        return dist;
    }
};

/**
 * Uso[1]: auto order = TopoSort<LL>().run(g);
 * Orden topologico de un grafo dirigido aciclico;
 * devuelve vacio si existe un ciclo.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct TopoSort {
    vi run(const Graph<T>& g) {
        vi in(g.n + 1, 0), order;
        for (auto& e : g.edges) in[e.to]++;
        queue<int> q;
        for (int i = 1; i <= g.n; i++) if (!in[i]) q.push(i);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            order.pb(u);
            for (int id : g.adj[u]) if (--in[g.edges[id].to] == 0) q.push(g.edges[id].to);
        }
        return order.size() == g.n ? order : vi();
    }
};

/**
 * Uso[1]: auto comp = TarjanSCC<LL>().run(g);
 * Componentes fuertemente conexas mediante Tarjan;
 * devuelve el id de componente de cada nodo.
 * Complejidad:  $O(V + E)$ .
 */
template <typename T = ll> struct TarjanSCC {
    vi run(const Graph<T>& g) {

```

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        int n = g.n, timer = 0, ncomps = 0;
        vi val(n + 1, 0), comp(n + 1, -1), stk;
        auto dfs = [&](auto&& self, int u) -> int {
            int low = val[u] = ++timer;
            stk.pb(u);
            for (int id : g.adj[u]) {
                int v = g.edges[id].to;
                if (comp[v] < 0) low = min(low, val[v] ? val[v] : self(self, v));
            }
            if (low == val[u]) {
                int x;
                do { x = stk.back(); stk.pop_back(); comp[x] = ncomps; } while (x != u);
                ncomps++;
            }
            return val[u] = low;
        };
        for (int i = 1; i <= n; i++) if (!val[i]) dfs(dfs, i);
        return comp;
    }
};

/**
 * Uso[1]: auto [sin_ciclo_neg, dist, parent, ciclo] =
 * BellmanFord<LL>().run(g, src);
 * Caminos minimos con pesos negativos; detecta y
 * recupera un ciclo negativo si existe.
 * Complejidad:  $O(V \cdot E)$ .
 */
template <typename T = ll> struct BellmanFord {
    tuple<bool, vector<T>, vi, vi> run(const Graph<T>& g,
        int src) {
        vector<T> dist(g.n + 1, LINF);
        vi p(g.n + 1, -1);
        dist[src] = 0;
        int x = -1;
        for (int i = 0; i < g.n; i++) {
            x = -1;
            for (auto& e : g.edges) if (dist[e.from] < LINF &&
                dist[e.to] > dist[e.from] + e.weight) {
                    dist[e.to] = max(-LINF, dist[e.from] + e.weight);
            }
            p[e.to] = e.from; x = e.to;
        }
        vi cycle;
        bool ok = true;
        if (x != -1) {
            ok = false;
            int y = x;
            for (int i = 0; i < g.n; i++) y = p[y];
            for (int cur = y; cur = p[cur]) { cycle.pb(cur);
                if (cur == y && cycle.size() > 1) break; }
            reverse(all(cycle));
        }
        return {ok, dist, p, cycle};
    }
};

/**
 * Uso[1]: auto ciclo = DirectedCycle<LL>().find(g);

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* Encuentra un ciclo en un grafo dirigido (vacío si no
  existe).
* Complejidad:  $O(V + E)$ .
*/
template <typename T = ll> struct DirectedCycle {
  vi find(const Graph<T>& g) {
    int n = g.n;
    vi state(n + 1, 0), parent(n + 1, -1), cycle;
    auto dfs = [&](auto&& self, int u) -> bool {
      state[u] = 1;
      for (int id : g.adj[u]) {
        int v = g.edges[id].to;
        if (state[v] == 0) {
          parent[v] = u;
          if (self(self, v)) return true;
        } else if (state[v] == 1) {
          cycle.pb(v);
          for (int cur = u; cur != -1 && cur != v; cur =
            parent[cur]) cycle.pb(cur);
          cycle.pb(v);
          reverse(all(cycle));
          return true;
        }
      }
      state[u] = 2;
      return false;
    };
    for (int i = 1; i <= n; i++) if (!state[i] && dfs(
      dfs, i)) return cycle;
    return cycle;
  }
};

/**
* Uso[1]: auto ciclo = UndirectedCycle<ll>().find(g);
* Encuentra un ciclo en un grafo no dirigido (vacío si
  no existe).
* Complejidad:  $O(V + E)$ .
*/
template <typename T = ll> struct UndirectedCycle {
  vi find(const Graph<T>& g) {
    int n = g.n;
    vi state(n + 1, 0), parent(n + 1, -1), cycle;
    auto dfs = [&](auto&& self, int u, int pe) -> bool {
      state[u] = 1;
      for (int id : g.adj[u]) {
        if ((id ^ 1) == pe) continue;
        int v = g.edges[id].to;
        if (state[v] == 0) {
          parent[v] = u;
          if (self(self, v, id)) return true;
        } else if (state[v] == 1) {
          cycle.pb(v);
          for (int cur = u; cur != -1 && cur != v; cur =
            parent[cur]) cycle.pb(cur);
          cycle.pb(v);
          reverse(all(cycle));
          return true;
        }
      }
      state[u] = 2;
      return false;
    };
  };
};

```

```

for (int i = 1; i <= n; i++) if (!state[i] && dfs(
  dfs, i, -1)) return cycle;
return cycle;
}

/**
* Uso[1]: auto [peso, aristas_mst] = Kruskal<ll>().run(
  g);
* Arbol de expansión mínima (MST) mediante Kruskal;
  devuelve el peso total y los índices de las aristas
* Complejidad:  $O(E \log E)$ .
*/
template <typename T = ll> struct Kruskal {
  pair<T, vi> run(const Graph<T>& g) {
    vi idx(g.edges.size());
    iota(all(idx), 0);
    sort(all(idx), [&](int a, int b) { return g.edges[a]
      .weight < g.edges[b].weight; });
    DSU dsu(g.n + 1);
    T total = 0;
    vi mst;
    for (int i : idx) {
      auto& e = g.edges[i];
      if (dsu.unite(e.from, e.to)) {
        total += e.weight;
        mst.pb(i);
        if (mst.size() == g.n - 1) break;
      }
    }
    return {total, mst};
  }
};

/**
* Uso[1]: auto circuito = EulerianCircuit<ll>().find(g,
  start, undirected);
* Circuito euleriano dirigido o no dirigido (vacío si
  no existe).
* Complejidad:  $O(V + E)$ .
*/
template <typename T = ll> struct EulerianCircuit {
  vi find(const Graph<T>& g, int start = 1, bool
    undirected = true) {
    int n = g.n;
    vi in(n + 1, 0), out(n + 1, 0);
    int max_id = -1;
    for (auto& e : g.edges) {
      out[e.from]++; in[e.to]++;
      max_id = max(max_id, e.id);
    }
    for (int i = 1; i <= n; i++) {
      if (undirected ? (out[i] & 1) : (in[i] != out[i]))
        return {};
    }
    vector<bool> used(max_id + 1, false);
    vi circuit, head(n + 1, 0);
    auto dfs = [&](auto&& self, int u) -> void {
      while (head[u] < g.adj[u].size()) {
        auto& e = g.edges[g.adj[u][head[u]++]];
        if (e.id == -1 || !used[e.id]) {
          if (e.id != -1) used[e.id] = true;

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          self(self, e.to);
        }
      }
      circuit.pb(u);
    };
    dfs(dfs, start);
    reverse(all(circuit));
    int need = undirected ? g.edges.size() / 2 + 1 : g.
      edges.size() + 1;
    if (circuit.size() != need && !g.edges.empty())
      return {};
    return circuit;
  }
};

/**
* Uso[1]: FunctionalGraph fg(n); fg.build_lifting(succ)
  ; fg.kth_successor(u, k); fg.decompose(succ);
* Grafo funcional (cada nodo un único sucesor): k-esimo
  sucesor con binary lifting y descomposición en
  ciclos.
* Complejidad: preproceso  $O(N \log K)$ , k-esimo sucesor
   $O(\log K)$ .
*/
struct FunctionalGraph {
  int n, log_k;
  vector<vector<int>> up;
  vi succ, in_cycle, cycle_id, cycle_pos, cycle_size,
    dist_to_cycle;
  vector<vi> cycles;
  FunctionalGraph(int n, int log_k = 60) : n(n), log_k(
    log_k),
    up(n + 1, vector<int>(log_k, 0)), succ(n + 1, 0),
    in_cycle(n + 1, 0),
    cycle_id(n + 1, -1), cycle_pos(n + 1, -1),
    cycle_size(n + 1, 0), dist_to_cycle(n + 1, 0) {}
  // Descompone el grafo en ciclos y arboles colgantes
  void decompose(const vi& _succ) {
    succ = _succ;
    vi state(n + 1, 0);
    cycles.clear();
    for (int i = 1; i <= n; i++) if (!state[i]) {
      int cur = i;
      vi path;
      while (cur >= 1 && cur <= n && state[cur] == 0) {
        state[cur] = 1;
        path.pb(cur);
        cur = succ[cur];
      }
      if (cur >= 1 && cur <= n && state[cur] == 1) {
        vi cyc;
        bool start = false;
        for (int u : path) { if (u == cur) start = true;
          if (start) cyc.pb(u); }
        int cid = cycles.size(), csz = cyc.size();
        for (int pos = 0; pos < csz; pos++) {
          int u = cyc[pos];
          in_cycle[u] = 1; cycle_id[u] = cid; cycle_pos[
            u] = pos; cycle_size[u] = csz;
          dist_to_cycle[u] = 0;
        }
        cycles.pb(cyc);
      }
    }
  }
};

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    for (int u : path) state[u] = 2;
}
for (int i = 1; i <= n; i++) if (!lin_cycle[i]) {
    int cur = i;
    vi path;
    while (cur >= 1 && cur <= n && lin_cycle[cur] &&
        cycle_id[cur] == -1) path.pb(cur), cur = succ[cur];
    int cid = (cur >= 1 && cur <= n) ? cycle_id[cur] :
        -1;
    int csz = (cur >= 1 && cur <= n) ? cycle_size[cur] :
        0;
    int d = (cur >= 1 && cur <= n) ? dist_to_cycle[cur] :
        0;
    for (int j = path.size() - 1; j >= 0; j--) {
        int u = path[j];
        cycle_id[u] = cid; cycle_size[u] = csz;
        dist_to_cycle[u] = ++d;
    }
}
// Binary Lifting sobre succ para responder k-esimo
// sucesor
void build_lifting(const vi& _succ) {
    succ = _succ;
    for (int i = 1; i <= n; i++) up[i][0] = succ[i];
    for (int j = 1; j < log_k; j++)
        for (int i = 1; i <= n; i++) {
            int p = up[i][j - 1];
            up[i][j] = (p >= 1 && p <= n) ? up[p][j - 1] :
                0;
        }
}
int kth_successor(int u, ll k) {
    for (int j = 0; j < log_k; j++) if (k & (1LL << j)) {
        u = up[u][j];
        if (u < 1 || u > n) return 0;
    }
    return u;
}
};

/**
 * Uso[1]: LCA lca(n); lca.build(root, adj); lca.lca(u,
 * v); lca.dist(u, v);
 * Ancestro comun mas bajo y distancia en numero de
 * aristas mediante binary lifting.
 * Complejidad: preproceso  $O(N \log N)$ , consulta  $O(\log N)$ .
 */
struct LCA {
    int n, log_n;
    vector<vi> up;
    vi depth;
    LCA(int n, int log_n = 20) : n(n), log_n(log_n), up(n
        + 1, vi(log_n, 0)), depth(n + 1, 0) {}
    void dfs(int u, int p, const vector<vi>& adj) {
        up[u][0] = p;
        for (int j = 1; j < log_n; j++) up[u][j] = up[up[u][
            j - 1]][j - 1];
        for (int v : adj[u]) if (v != p) depth[v] = depth[u]
            + 1, dfs(v, u, adj);
    }
    void build(int root, const vector<vi>& adj) { dfs(root

```

```

        , root, adj); }
    int lca(int u, int v) {
        if (depth[u] < depth[v]) swap(u, v);
        int k = depth[u] - depth[v];
        for (int j = 0; j < log_n; j++) if (k & (1 << j)) u
            = up[u][j];
        if (u == v) return u;
        for (int j = log_n - 1; j >= 0; j--) if (up[u][j] !=
            up[v][j]) u = up[u][j], v = up[v][j];
        return up[u][0];
    }
    int dist(int u, int v) { return depth[u] + depth[v] -
        2 * depth[lca(u, v)]; }
};

/**
 * Uso[1]: TwoSat s(n); s.add_clause(u, tu, v, tv); s.
 * force_value(u, t); s.solve();
 * Satisfacibilidad booleana 2-SAT con clausulas (or,
 * implicacion, xor, equivalencia) y la asignacion
 * resultante.
 * Complejidad:  $O(V + E)$ .
 */
struct TwoSat {
    int n;
    Graph<ll> G;
    vector<bool> assignment;
    TwoSat(int n) : n(n), G(2 * n), assignment(n + 1,
        false) {}
    int node(int u, bool t) { return t ? u : u + n; }
    void add_clause(int u, bool tu, int v, bool tv) {
        G.add_directed_edge(node(u, !tu), node(v, tv));
        G.add_directed_edge(node(v, !tv), node(u, tu));
    }
    void force_value(int u, bool t) { add_clause(u, t, u,
        t); }
    void add_implication(int u, bool tu, int v, bool tv) {
        add_clause(u, !tu, v, tv); }
    void add_xor(int u, bool tu, int v, bool tv) {
        add_clause(u, tu, v, tv);
        add_clause(u, !tu, v, !tv);
    }
    void add_equivalence(int u, bool tu, int v, bool tv) {
        add_implication(u, tu, v, tv);
        add_implication(v, tv, u, tu);
    }
    bool solve() {
        vi comp = TarjanSCC<ll>().run(G);
        for (int i = 1; i <= n; i++) {
            if (comp[i] == comp[i + n]) return false;
            assignment[i] = comp[i] < comp[i + n];
        }
        return true;
    }
};

/**
 * Uso[1]: BipartiteMatcher bm(n_izq, n_der); bm.
 * add_edge(u, v); bm.max_matching();
 * Emparejamiento maximo bipartito con el algoritmo de
 * Kuhn.
 * Complejidad:  $O(V \cdot E)$ .
 */

```

```

struct BipartiteMatcher {
    int n, m;
    vector<vi> adj;
    vi match;
    BipartiteMatcher(int n, int m) : n(n), m(m), adj(n +
        1), match(m + 1, -1) {}
    void add_edge(int u, int v) { adj[u].pb(v); }
    bool dfs(int u, vector<bool>& vis) {
        for (int v : adj[u]) if (!vis[v]) {
            vis[v] = true;
            if (match[v] < 0 || dfs(match[v], vis)) { match[v]
                = u; return true; }
        }
        return false;
    }
    int max_matching() {
        int ans = 0;
        for (int i = 1; i <= n; i++) {
            vector<bool> vis(m + 1, false);
            if (dfs(i, vis)) ans++;
        }
        return ans;
    }
};

/**
 * Uso[1]: HopcroftKarp hk(n_izq, n_der); hk.add_edge(u,
 * v); hk.max_matching();
 * Emparejamiento maximo bipartito con el algoritmo de
 * Hopcroft-Karp.
 * Complejidad:  $O(E\sqrt{V})$ .
 */
struct HopcroftKarp {
    int n, m;
    vector<vi> adj;
    vi pairU, pairV, dist;
    HopcroftKarp(int n, int m) : n(n), m(m), adj(n + 1),
        pairU(n + 1, 0), pairV(m + 1, 0), dist(n + 1) {}
    void add_edge(int u, int v) { adj[u].pb(v); }
    bool bfs() {
        queue<int> q;
        for (int u = 1; u <= n; u++) {
            if (pairU[u]) dist[u] = INF;
            else dist[u] = 0, q.push(u);
        }
        dist[0] = INF;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            if (dist[u] < dist[0])
                for (int v : adj[u]) if (dist[pairV[v]] == INF)
                    dist[pairV[v]] = dist[u] + 1, q.push(pairV[v]);
        }
        return dist[0] != INF;
    }
    bool dfs(int u) {
        if (!u) return true;
        for (int v : adj[u]) if (dist[pairV[v]] == dist[u] +
            1 && dfs(pairV[v])) {
            pairV[v] = u; pairU[u] = v; return true;
        }
        dist[u] = INF;
        return false;
    }
}

```

```

int max_matching() {
    int res = 0;
    while (bfs()) for (int u = 1; u <= n; u++) if (!
        pairU[u] && dfs(u)) res++;
    return res;
}
};
/**
 * Uso[0]: MaxClique mc(n); mc.add_edge(u, v); mc.solve
    ();
 * Clique maxima de un grafo con hasta 60 nodos usando
    bitsets.
 * Complejidad: exponencial en el peor caso, practico
    para  $N \leq 60$ .
 */
struct MaxClique {
    int n, max_sz;
    vector<ll> adj;
    MaxClique(int n) : n(n), max_sz(0), adj(n, 0) {}
    void add_edge(int u, int v) { adj[u] |= (1LL << v);
        adj[v] |= (1LL << u); }
    void dfs(ll R, ll P, ll X) {
        if (!P && !X) { max_sz = max(max_sz,
            __builtin_popcountll(R)); return; }
        ll iter = P & ~adj[__builtin_ctzll(P | X)];
        while (iter) {
            int v = __builtin_ctzll(iter); ll v_bit = 1LL << v
            ;
            dfs(R | v_bit, P & adj[v], X & adj[v]);
            P &= ~v_bit; X |= v_bit; iter &= ~v_bit;
        }
    }
    int solve() { dfs(0, (1LL << n) - 1, 0); return max_sz
    ; }
};
/**
 * Uso[1]: BridgeTree bt(n, num_aristas); bt.add_edge(u,
    v, id); bt.build();
 * Detecta puentes y puntos de articulacion y construye
    el arbol de componentes 2-arista-conexas.
 * Complejidad:  $O(V + E)$ .
 */
struct BridgeTree {
    int n, timer, num_comps;
    vector<vector<pair<int, int>>> adj;
    vi tin, low, comp_id;
    vector<bool> is_bridge, is_articulation;
    vector<vi> tree_adj;
    BridgeTree(int n, int num_edges) : n(n), adj(n + 1),
        tin(n + 1, -1), low(n + 1, -1),
        comp_id(n + 1, 0), is_bridge(num_edges, false),
        is_articulation(n + 1, false), timer(0), num_comps
        (0) {}
    void add_edge(int u, int v, int id) { adj[u].pb({v, id
        }); adj[v].pb({u, id}); }
    void dfs_tarjan(int u, int p = -1) {
        tin[u] = low[u] = ++timer;
        int children = 0;
        for (auto [v, id] : adj[u]) {
            if (v == p) continue;
            if (tin[v] != -1) low[u] = min(low[u], tin[v]);
            else {

```

```

                children++;
                dfs_tarjan(v, u);
                low[u] = min(low[u], low[v]);
                if (low[v] > tin[u]) is_bridge[id] = true;
                if (low[v] >= tin[u] && p != -1) is_articulation
                    [u] = true;
            }
        }
        if (p == -1 && children > 1) is_articulation[u] =
            true;
    }
    void dfs_comp(int u, int cur) {
        comp_id[u] = cur;
        for (auto [v, id] : adj[u]) if (!comp_id[v]) {
            if (is_bridge[id]) {
                tree_adj.pb(vi());
                tree_adj[cur].pb(++num_comps);
                tree_adj[num_comps].pb(cur);
                dfs_comp(v, num_comps);
            } else dfs_comp(v, cur);
        }
    }
    void build() {
        for (int i = 1; i <= n; i++) if (tin[i] == -1)
            dfs_tarjan(i);
        tree_adj.pb(vi());
        for (int i = 1; i <= n; i++) if (!comp_id[i])
            tree_adj.pb(vi()), dfs_comp(i, ++num_comps);
    }
};
/**
 * Uso[0]: Dinic d(n, s, t); d.add_edge(u, v, cap); d.
    max_flow();
 * Flujo maximo con el algoritmo de Dinic, dirigido o no
    dirigido.
 * Complejidad:  $O(V^2E)$  en general,  $O(E\sqrt{V})$  en redes
    unitarias.
 */
struct Dinic {
    struct FlowEdge { int u, v; ll cap, flow = 0; };
    int n, s, t;
    vector<FlowEdge> edges;
    vector<vi> adj;
    vi level, ptr;
    Dinic(int n, int s, int t) : n(n), s(s), t(t), adj(n +
        1), level(n + 1), ptr(n + 1) {}
    int add_edge(int u, int v, ll cap, bool directed =
        true) {
        adj[u].pb(edges.size()); edges.pb({u, v, cap});
        adj[v].pb(edges.size()); edges.pb({v, u, directed ?
            0 : cap});
        return (int)edges.size() - 2; // indice de la arista
            forward
    }
    bool bfs() {
        fill(all(level), -1);
        level[s] = 0;
        queue<int> q;
        q.push(s);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int id : adj[u]) {

```

```

                auto& e = edges[id];
                if (e.cap - e.flow >= 1 && level[e.v] == -1)
                    level[e.v] = level[u] + 1, q.push(e.v);
            }
        }
        return level[t] != -1;
    }
    ll dfs(int u, ll pushed) {
        if (!pushed) return 0;
        if (u == t) return pushed;
        for (int& cid = ptr[u]; cid < adj[u].size(); cid++)
        {
            auto& e = edges[adj[u][cid]];
            ll tr = e.cap - e.flow;
            if (level[u] + 1 != level[e.v] || !tr) continue;
            ll push = dfs(e.v, min(pushed, tr));
            if (push) {
                e.flow += push;
                edges[adj[u][cid] ^ 1].flow -= push;
                return push;
            }
        }
        return 0;
    }
    ll max_flow() { return max_flow(s, t); }
    ll max_flow(int src, int snk) {
        s = src; t = snk;
        ll flow = 0;
        while (bfs()) {
            fill(all(ptr), 0);
            while (ll pushed = dfs(s, LINF)) flow += pushed;
        }
        return flow;
    }
};
/**
 * Uso[1]: DinicLB d(n); int id = d.add_edge(u, v, lo,
    hi); ll flujo = d.max_flow(s, t); ll f = d.
    edge_flow(id);
 * Flujo maximo con cotas inferiores (Lower bounds) en
    cada arista; devuelve -1 si no existe flujo
    factible.
 * Complejidad:  $O(\text{Dinic})$  sobre el grafo transformado (
    dos fases de flujo maximo).
 */
struct DinicLB {
    int n, ss, tt;
    Dinic din;
    vector<ll> bal, lo;
    vector<int> edge_idx;
    bool built = false;
    ll total = 0;
    DinicLB(int n) : n(n), ss(0), tt(n + 1), din(n + 2, 0,
        0), bal(n + 2, 0) {}
    int add_edge(int u, int v, ll lower, ll upper) {
        bal[u] -= lower; bal[v] += lower;
        lo.pb(lower);
        int id = din.add_edge(u, v, upper - lower);
        edge_idx.pb(id);
        return (int)lo.size() - 1;
    }
    void build_super() {

```

```

    if (built) return;
    built = true;
    for (int i = 1; i <= n; i++) {
        if (bal[i] > 0) din.add_edge(ss, i, bal[i]), total
            += bal[i];
        else if (bal[i] < 0) din.add_edge(i, tt, -bal[i]);
    }
}
bool feasible() {
    build_super();
    return din.max_flow(ss, tt) == total;
}
ll max_flow(int s, int t) {
    int idx = din.add_edge(t, s, LINF);
    if (!feasible()) return -1;
    ll flow = din.edges[idx].flow; // flujo de la
    // eliminacion (valor base s->t)
    // elimina t->s (cap y flow en ambas direcciones, si
    // no queda flujo residual cancelable)
    din.edges[idx].flow = 0; din.edges[idx].cap = 0;
    din.edges[idx ^ 1].flow = 0; din.edges[idx ^ 1].cap
        = 0;
    return flow + din.max_flow(s, t);
}
ll edge_flow(int id) const { return lo[id] + din.edges
    [edge_idx[id]].flow; }
};

/**
 * Uso[0]: MCMF m(n); m.add_edge(u, v, cap, costo); auto
    [flujo, costo] = m.solve(s, t);
 * Flujo maximo de costo minimo (min cost max flow) con
    potenciales para costos negativos.
 * Complejidad:  $O(F \cdot E \log V)$  con  $F$  el flujo total.
 */
struct MCMF {
    struct Edge { int to, rev; ll cap, flow, cost; };
    int n;
    vector<vector<Edge>> adj;
    vll dist, pot;
    vi p_node, p_edge;
    MCMF(int n) : n(n), adj(n + 1), dist(n + 1), pot(n +
        1, 0), p_node(n + 1), p_edge(n + 1) {}
    void add_edge(int u, int v, ll cap, ll cost) {
        adj[u].pb({v, (int)adj[v].size(), cap, 0, cost});
        adj[v].pb({u, (int)adj[u].size() - 1, 0, 0, -cost});
    }
    // Inicializa potenciales si hay costos negativos.
    // Omitir si todos son >= 0.
    void init_potentials(int s) {
        fill(all(pot), LINF);
        pot[s] = 0;
        for (int i = 0; i < n; i++)
            for (int u = 0; u <= n; u++) if (pot[u] != LINF)
                for (auto& e : adj[u]) if (e.cap - e.flow > 0 &&
                    pot[e.to] > pot[u] + e.cost) pot[e.to] = pot[u] +
                    e.cost;
    }
    bool dijkstra(int s, int t) {
        fill(all(dist), LINF);
        dist[s] = 0;
        priority_queue<pair<ll, int>, vector<pair<ll, int>>,
            greater<>> pq;

```

```

        pq.push({0, s});
        while (!pq.empty()) {
            auto [d, u] = pq.top(); pq.pop();
            if (d > dist[u]) continue;
            for (int i = 0; i < adj[u].size(); i++) {
                auto& e = adj[u][i];
                ll rc = e.cost + pot[u] - pot[e.to]; // costo
                reducido
                if (e.cap - e.flow > 0 && dist[e.to] > dist[u] +
                    rc) {
                    dist[e.to] = dist[u] + rc;
                    p_node[e.to] = u; p_edge[e.to] = i;
                    pq.push({dist[e.to], e.to});
                }
            }
            return dist[t] != LINF;
        }
    }
    pair<ll, ll> solve(int s, int t) { // {flujo, costo}
        init_potentials(s);
        ll flow = 0, cost = 0;
        while (dijkstra(s, t)) {
            for (int i = 0; i <= n; i++) if (dist[i] != LINF)
                pot[i] += dist[i];
            ll push = LINF;
            for (int cur = t; cur != s; cur = p_node[cur])
                push = min(push, adj[p_node[cur]][p_edge[cur]].
                    cap - adj[p_node[cur]][p_edge[cur]].flow);
            for (int cur = t; cur != s; cur = p_node[cur]) {
                auto& e = adj[p_node[cur]][p_edge[cur]];
                e.flow += push;
                adj[cur][e.rev].flow -= push;
                cost += push * e.cost;
            }
            flow += push;
        }
        return {flow, cost};
    }
};

/**
 * Uso[1]: DominatorTree dt(n); dt.add_edge(u, v); dt.
    build(root);
 * Arbol dominador de un grafo dirigido desde una raiz;
    dom[u] son los hijos de u en el arbol.
 * Complejidad:  $O((V + E) \log V)$ .
 */
struct DominatorTree {
    int n, t;
    vector<vi> adj, rev, dom;
    vi dfn, id, p, sdom, idom, dsu, best;
    DominatorTree(int n) : n(n), t(0), adj(n + 1), rev(n +
        1), dom(n + 1),
        dfn(n + 1, 0), id(n + 1), p(n + 1), sdom(n + 1),
        idom(n + 1), dsu(n + 1), best(n + 1) {}
    void add_edge(int u, int v) { adj[u].pb(v); }
    void dfs(int u) {
        dfn[u] = ++t;
        id[t] = u;
        sdom[t] = best[t] = dsu[t] = t;
        for (int v : adj[u]) {
            if (!dfn[v]) dfs(v), p[dfn[v]] = dfn[u];
            rev[dfn[v]].pb(dfn[u]);
        }
    }

```

```

    }
}
int find(int x) {
    if (x == dsu[x]) return x;
    int y = find(dsu[x]);
    if (sdom[best[x]] > sdom[best[dsu[x]]]) best[x] =
        best[dsu[x]];
    return dsu[x] = y;
}
void build(int root) {
    dfs(root);
    vector<vi> bucket(n + 1);
    for (int i = t; i >= 2; i--) {
        for (int u : rev[i]) {
            find(u);
            if (sdom[best[u]] < sdom[i]) sdom[i] = sdom[best
                [u]];
        }
        bucket[sdom[i]].pb(i);
        dsu[i] = p[i];
        for (int u : bucket[p[i]]) {
            find(u);
            idom[u] = sdom[best[u]] == sdom[u] ? sdom[u] :
                best[u];
        }
        bucket[p[i]].clear();
    }
    for (int i = 2; i <= t; i++) {
        if (idom[i] != sdom[i]) idom[i] = idom[idom[i]];
        dom[id[idom[i]]].pb(id[i]);
    }
}
};

```

Trees

```

/**
 * Uso[1]: TreeEdge<T> e = {to, id, weight};
 * Arista de arbol con destino, identificador opcional y
    peso generico.
 * Complejidad:  $O(1)$  por arista.
 */
template <typename T = ll> struct TreeEdge { int to, id;
    T weight; };

/**
 * Uso[1]: Tree<LL> t(n); t.add_edge(u, v, w); t.init(
    root); t.get_diameter(); t.get_centers(); t.
    all_farthest_distances();
 * Arbol con raiz: tiempos de entrada/salida (euler),
    tamanos de subarbol, diametro con camino y centros.
 * Complejidad: init y diametro  $O(N)$ .
 */
template <typename T = ll> struct Tree {
    int n, root, timer;
    vector<vector<TreeEdge<T>>> adj;
    vi parent, depth, sz, tin, tout, euler_order;
    vector<T> dist_from_root;
    Tree(int n = 0) : n(n), root(1), timer(0), adj(n + 1),
        parent(n + 1), depth(n + 1),
        sz(n + 1), tin(n + 1), tout(n + 1), dist_from_root(n
            + 1) {}
    void add_edge(int u, int v, T w = 1, int id = -1) {

```

```

    adj[u].pb({v, id, w});
    adj[v].pb({u, id, w});
}
void add_directed_edge(int u, int v, T w = 1, int id = -1) { adj[u].pb({v, id, w}); }
void dfs_init(int u, int p = 0, int d = 0, T cur = 0)
{
    parent[u] = p; depth[u] = d; dist_from_root[u] = cur;
    sz[u] = 1; tin[u] = ++timer;
    euler_order.pb(u);
    for (auto& e : adj[u]) if (e.to != p) {
        dfs_init(e.to, u, d + 1, cur + e.weight);
        sz[u] += sz[e.to];
    }
    tout[u] = timer;
}
void init(int root = 1) {
    this->root = root; timer = 0;
    euler_order.clear();
    dfs_init(root, 0, 0, 0);
}
struct DiameterResult { T length; int u, v; vi path; };
DiameterResult get_diameter() {
    auto farthest = [&](int start) {
        vector<T> dist(n + 1, -1);
        queue<int> q;
        q.push(start); dist[start] = 0;
        int far = start;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            if (dist[u] > dist[far]) far = u;
            for (auto& e : adj[u]) if (dist[e.to] == -1)
                dist[e.to] = dist[u] + e.weight, q.push(e.to);
        }
        return pair<int, T>(far, dist[far]);
    };
    int u = farthest(1).fi;
    auto [v, length] = farthest(u);
    vi prev(n + 1, -1);
    queue<int> q;
    q.push(u); prev[u] = 0;
    while (!q.empty()) {
        int cur = q.front(); q.pop();
        if (cur == v) break;
        for (auto& e : adj[cur]) if (prev[e.to] == -1)
            prev[e.to] = cur, q.push(e.to);
    }
    vi path;
    for (int cur = v; cur != 0; cur = prev[cur]) path.pb(cur);
    reverse(all(path));
    return {length, u, v, path};
}
// 1 o 2 nodos centrales del diametro
vi get_centers() {
    auto d = get_diameter();
    int m = d.path.size();
    if (m & 1) return {d.path[m / 2]};
    return {d.path[m / 2 - 1], d.path[m / 2]};
}
// Distancia maxima desde cada nodo a cualquier otro.

```

```

O(N)
vector<T> all_farthest_distances() {
    auto d = get_diameter();
    auto dist_from = [&](int start) {
        vector<T> dist(n + 1, -1);
        queue<int> q;
        q.push(start); dist[start] = 0;
        while (!q.empty()) {
            int cur = q.front(); q.pop();
            for (auto& e : adj[cur]) if (dist[e.to] == -1)
                dist[e.to] = dist[cur] + e.weight, q.push(e.to);
        }
        return dist;
    };
    auto du = dist_from(d.u), dv = dist_from(d.v);
    vector<T> res(n + 1);
    for (int i = 1; i <= n; i++) res[i] = max(du[i], dv[i]);
    return res;
}
/**
 * Uso[1]: BinaryLifting<LL> bl(t, root); bl.lca(u, v);
 * bl.kth_ancestor(u, k); bl.path_sum(u, v); bl.
 * dist_weighted(u, v);
 * Binary lifting sobre un arbol: lca, k-esimo ancestro,
 * suma/max/min de aristas en el camino y distancias.
 * Complejidad: preproceso O(N log N), consulta O(log N).
 */
template <typename T = ll> struct BinaryLifting {
    int n, log_n;
    vi depth;
    vector<vi> up;
    vector<vector<T>> mx, mn, sum;
    vector<T> dist_root;
    BinaryLifting(const Tree<T>& tree, int root = 1) {
        n = tree.n;
        log_n = 32 - __builtin_clz(n) + 1;
        depth = tree.depth;
        dist_root = tree.dist_from_root;
        up.assign(n + 1, vi(log_n, 0));
        mx.assign(n + 1, vector<T>(log_n, -LINF));
        mn.assign(n + 1, vector<T>(log_n, LINF));
        sum.assign(n + 1, vector<T>(log_n, 0));
        for (int u = 1; u <= n; u++) up[u][0] = tree.parent[u] ? tree.parent[u] : u;
        for (int u = 1; u <= n; u++)
            for (auto& e : tree.adj[u]) if (e.to == tree.parent[u])
                mx[u][0] = mn[u][0] = sum[u][0] = e.weight;
        for (int j = 1; j < log_n; j++)
            for (int i = 1; i <= n; i++) {
                int p = up[i][j - 1];
                up[i][j] = up[p][j - 1];
                mx[i][j] = max(mx[i][j - 1], mx[p][j - 1]);
                mn[i][j] = min(mn[i][j - 1], mn[p][j - 1]);
                sum[i][j] = sum[i][j - 1] + sum[p][j - 1];
            }
    }
    int kth_ancestor(int u, int k) {
        if (depth[u] < k) return 0;
        for (int j = 0; j < log_n; j++) if (k & (1 << j)) u

```

```

        = up[u][j];
        return u;
    }
    int lca(int u, int v) {
        if (depth[u] < depth[v]) swap(u, v);
        u = kth_ancestor(u, depth[u] - depth[v]);
        if (u == v) return u;
        for (int j = log_n - 1; j >= 0; j--) if (up[u][j] != up[v][j]) u = up[u][j], v = up[v][j];
        return up[u][0];
    }
    int dist_edges(int u, int v) { return depth[u] + depth[v] - 2 * depth[lca(u, v)]; }
    T dist_weighted(int u, int v) { return dist_root[u] + dist_root[v] - 2 * dist_root[lca(u, v)]; }
    // k-esimo nodo en el camino u->v (0-indexed: 0 = u)
    int kth_node_on_path(int u, int v, int k) {
        int anc = lca(u, v);
        int d1 = depth[u] - depth[anc], d2 = depth[v] - depth[anc];
        if (k <= d1) return kth_ancestor(u, k);
        k -= d1;
        if (k <= d2) return kth_ancestor(v, d2 - k);
        return 0;
    }
    T path_max(int u, int v) {
        int anc = lca(u, v);
        T res = -LINF;
        auto lift = [&](int node, int steps) {
            for (int j = 0; j < log_n; j++) if (steps & (1 << j)) res = max(res, mx[node][j]), node = up[node][j];
        };
        lift(u, depth[u] - depth[anc]);
        lift(v, depth[v] - depth[anc]);
        return res;
    }
    T path_min(int u, int v) {
        int anc = lca(u, v);
        T res = LINF;
        auto lift = [&](int node, int steps) {
            for (int j = 0; j < log_n; j++) if (steps & (1 << j)) res = min(res, mn[node][j]), node = up[node][j];
        };
        lift(u, depth[u] - depth[anc]);
        lift(v, depth[v] - depth[anc]);
        return res;
    }
    T path_sum(int u, int v) {
        int anc = lca(u, v);
        T res = 0;
        auto lift = [&](int node, int steps) {
            for (int j = 0; j < log_n; j++) if (steps & (1 << j)) res += sum[node][j], node = up[node][j];
        };
        lift(u, depth[u] - depth[anc]);
        lift(v, depth[v] - depth[anc]);
        return res;
    }
    bool is_ancestor(int u, int v) { return lca(u, v) == u; }
};

```

```

/**
 * Uso[1]: EulerTourTree<ll> et(t); et.subtree_range(u);
 *           et.subtree_size(u);
 * Mapea el subarbol de cada nodo a un rango contiguo de
 * la secuencia de entrada (tin/tout).
 * Complejidad: O(1) por consulta.
 */
template <typename T = ll> struct EulerTourTree {
    vi in, out;
    EulerTourTree(const Tree<T>& tree) : in(tree.tin), out
    (tree.tout) {}
    pair<int, int> subtree_range(int u) { return {in[u],
    out[u]}; }
    int subtree_size(int u) { return out[u] - in[u] + 1; }
};

/**
 * Uso[1]: HLD<ll> hld(n); hld.add_edge(u, v); hld.init(
 * root); hld.update_path(u, v, val); hld.query_path(u
 * , v);
 * Heavy-Light decomposition con segment tree perezoso
 * para sumas y updates sobre caminos y subarboles.
 * Complejidad: preproceso O(N), update y query de
 * camino O(log2 N).
 */
template <typename T = ll> struct HLD {
    struct SegTree {
        int size;
        vector<T> tree, lazy;
        void init(int n) {
            size = 1;
            while (size < n) size <= 1;
            tree.assign(2 * size, 0);
            lazy.assign(2 * size, 0);
        }
        void push(int node, int l, int r) {
            if (!lazy[node]) return;
            tree[node] += lazy[node] * (r - l + 1);
            if (l != r) lazy[2 * node] += lazy[node], lazy[2 *
            node + 1] += lazy[node];
            lazy[node] = 0;
        }
        void update(int node, int l, int r, int ql, int qr,
        T val) {
            push(node, l, r);
            if (ql > r || qr < l) return;
            if (ql <= l && r <= qr) { lazy[node] += val; push(
            node, l, r); return; }
            int mid = (l + r) / 2;
            update(2 * node, l, mid, ql, qr, val);
            update(2 * node + 1, mid + 1, r, ql, qr, val);
            tree[node] = tree[2 * node] + tree[2 * node + 1];
        }
        T query(int node, int l, int r, int ql, int qr) {
            push(node, l, r);
            if (ql > r || qr < l) return 0;
            if (ql <= l && r <= qr) return tree[node];
            int mid = (l + r) / 2;
            return query(2 * node, l, mid, ql, qr) + query(2 *
            node + 1, mid + 1, r, ql, qr);
        }
    };
    seg;

```

```

    int n, cur_pos;
    vector<vi> adj;
    vi parent, depth, heavy, head, pos;
    HLD(int n) : n(n), cur_pos(1), adj(n + 1), parent(n +
    1), depth(n + 1),
        heavy(n + 1, -1), head(n + 1), pos(n + 1) { seg.init
        (n + 1); }
    void add_edge(int u, int v) { adj[u].pb(v); adj[v].pb(
    u); }
    int dfs_size(int u, int p, int d) {
        int sz = 1, max_c = 0;
        depth[u] = d; parent[u] = p; heavy[u] = -1;
        for (int v : adj[u]) if (v != p) {
            int cs = dfs_size(v, u, d + 1);
            sz += cs;
            if (cs > max_c) max_c = cs, heavy[u] = v;
        }
        return sz;
    }
    void decompose(int u, int h) {
        head[u] = h;
        pos[u] = cur_pos++;
        if (heavy[u] != -1) decompose(heavy[u], h);
        for (int v : adj[u]) if (v != parent[u] && v !=
        heavy[u]) decompose(v, v);
    }
    void init(int root = 1) { cur_pos = 1; dfs_size(root,
    0, 0); decompose(root, root); }
    void update_path(int u, int v, T val) {
        while (head[u] != head[v]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            seg.update(1, 1, n, pos[head[v]], pos[v], val);
            v = parent[head[v]];
        }
        if (depth[u] > depth[v]) swap(u, v);
        seg.update(1, 1, n, pos[u], pos[v], val);
    }
    T query_path(int u, int v) {
        T res = 0;
        while (head[u] != head[v]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            res += seg.query(1, 1, n, pos[head[v]], pos[v]);
            v = parent[head[v]];
        }
        if (depth[u] > depth[v]) swap(u, v);
        return res + seg.query(1, 1, n, pos[u], pos[v]);
    }
    void update_subtree(int u, int sz_u, T val) { seg.
    update(1, 1, n, pos[u], pos[u] + sz_u - 1, val); }
    T query_subtree(int u, int sz_u) { return seg.query(1,
    1, n, pos[u], pos[u] + sz_u - 1); }
    void update_node(int u, T val) { seg.update(1, 1, n,
    pos[u], pos[u], val); }
    int lca(int u, int v) {
        while (head[u] != head[v]) {
            if (depth[head[u]] > depth[head[v]]) swap(u, v);
            v = parent[head[v]];
        }
        return depth[u] < depth[v] ? u : v;
    }
};
/**

```

```

 * Uso[1]: CentroidDecomposition cd(n); cd.add_edge(u, v
 * , w); cd.K = lim; cd.init(true); cd.total_paths;
 * Descomposicion en centroides: conteo offline de pares
 * con distancia menor o igual a K y consultas online
 * de nodo mas cercano.
 * Complejidad: construccion O(N log N), update y query
 * online O(log N).
 */
struct CentroidDecomposition {
    int n;
    vector<vector<pair<int, ll>>> adj;
    vi sz, c_parent;
    vector<bool> removed;
    ll total_paths = 0, K = 0;
    vll local_ans;
    CentroidDecomposition(int n) : n(n), adj(n + 1), sz(n
    + 1), c_parent(n + 1),
        removed(n + 1, false), local_ans(n + 1, LINF) {}
    void add_edge(int u, int v, ll w = 1) { adj[u].pb({v,
    w}); adj[v].pb({u, w}); }
    int get_sizes(int u, int p = 0) {
        sz[u] = 1;
        for (auto [v, w] : adj[u]) if (v != p && !removed[v
        ]) sz[u] += get_sizes(v, u);
        return sz[u];
    }
    int get_centroid(int u, int p, int size) {
        for (auto [v, w] : adj[u]) if (v != p && !removed[v]
        && sz[v] > size / 2) return get_centroid(v, u,
        size);
        return u;
    }
    void get_paths(int u, int p, ll dist, vll& paths) {
        paths.pb(dist);
        for (auto [v, w] : adj[u]) if (v != p && !removed[v
        ]) get_paths(v, u, dist + w, paths);
    }
    ll count_pairs(vll& paths, ll limit) {
        sort(all(paths));
        ll cnt = 0;
        for (int l = 0, r = paths.size() - 1; l < r; ) {
            if (paths[l] + paths[r] <= limit) cnt += r - l, l
            ++;
            else r--;
        }
        return cnt;
    }
    void process_centroid(int c) {
        vll all_paths{0};
        for (auto [v, w] : adj[c]) if (!removed[v])
            get_paths(v, c, w, all_paths);
        total_paths += count_pairs(all_paths, K);
        for (auto [v, w] : adj[c]) if (!removed[v]) {
            vll sub;
            get_paths(v, c, w, sub);
            total_paths -= count_pairs(sub, K);
        }
    }
    int build_tree(int u, int p = 0, bool offline_mode =
    false) {
        int c = get_centroid(u, 0, get_sizes(u, 0));
        if (offline_mode) process_centroid(c);
        removed[c] = true;
    }

```

```

    c_parent[c] = p;
    for (auto [v, w] : adj[c]) if (!removed[v])
        build_tree(v, c, offline_mode);
    return c;
}
void init(bool offline_mode = false) { build_tree(1,
    0, offline_mode); }
// Pinta u de rojo. LCA del arbol original.
void update(int u, LCA& lca) {
    for (int cur = u; cur; cur = c_parent[cur])
        local_ans[cur] = min(local_ans[cur], (ll)lca.dist(u,
            cur));
}
ll query(int u, LCA& lca) {
    ll best = LINF;
    for (int cur = u; cur; cur = c_parent[cur]) best =
        min(best, (ll)lca.dist(u, cur) + local_ans[cur]);
    return best;
}
};

/**
 * Uso[1]: DSUOnTree s(n); s.add_edge(u, v); s.
 * set_colors(color_index); s.solve(root); s.ans[u];
 * Small-to-Large (sack) para responder estadísticas de
 * colores en cada subarbol.
 * Complejidad:  $O(N \log N)$ .
 */
struct DSUOnTree {
    int n, timer, distinct;
    vector<vi> adj;
    vi sz, heavy, tin, tout, euler_node, color, ans, cnt;
    DSUOnTree(int n) : n(n), timer(0), distinct(0), adj(n
        + 1), sz(n + 1), heavy(n + 1, -1),
        tin(n + 1), tout(n + 1), euler_node(n + 1), color(n
        + 1), ans(n + 1), cnt(n + 1) {}
    void add_edge(int u, int v) { adj[u].pb(v); adj[v].pb(
        u); }
    void set_colors(const vi& c) {
        for (int i = 1; i <= n; i++) color[i] = c[i];
    }
    void dfs_size(int u, int p = 0) {
        sz[u] = 1;
        tin[u] = ++timer;
        euler_node[timer] = u;
        int max_sz = 0;
        for (int v : adj[u]) if (v != p) {
            dfs_size(v, u);
            sz[u] += sz[v];
            if (sz[v] > max_sz) max_sz = sz[v], heavy[u] = v;
        }
        tout[u] = timer;
    }
    void add_node(int u) { if (!cnt[color[u]]++) distinct
        ++; }
    void remove_node(int u) { if (--cnt[color[u]])
        distinct--; }
    void dfs_sack(int u, int p, bool keep) {
        for (int v : adj[u]) if (v != p && v != heavy[u])
            dfs_sack(v, u, false);
        if (heavy[u] != -1) dfs_sack(heavy[u], u, true);
        add_node(u);
        for (int v : adj[u]) if (v != p && v != heavy[u])

```

```

        for (int t = tin[v]; t <= tout[v]; t++) add_node(
            euler_node[t]);
        ans[u] = distinct;
        if (!keep) for (int t = tin[u]; t <= tout[u]; t++)
            remove_node(euler_node[t]);
    }
    void solve(int root = 1) { timer = 0; dfs_size(root);
        dfs_sack(root, 0, false); }
};

/**
 * Uso[1]: VirtualTree vt(t, bl); auto [raiz, nodos] =
 * vt.build(vector_de_nodos);
 * Construye el arbol virtual sobre un subconjunto de
 * nodos, con aristas pesadas por distancia.
 * Complejidad:  $O(K \log N)$  con  $K$  el numero de nodos
 * solicitados.
 */
struct VirtualTree {
    const Tree<ll>& tree;
    BinaryLifting<ll>& bl;
    vector<vector<pair<int, ll>>> virt_adj;
    VirtualTree(const Tree<ll>& tree, BinaryLifting<ll>&
        bl) : tree(tree), bl(bl), virt_adj(tree.n + 1) {}
    // Retorna {raiz, nodos del arbol virtual}
    pair<int, vi> build(vi nodes) {
        sort(all(nodes), [&](int a, int b) { return tree.tin
            [a] < tree.tin[b]; });
        int k = nodes.size();
        for (int i = 0; i + 1 < k; i++) nodes.pb(bl.lca(
            nodes[i], nodes[i + 1]));
        sort(all(nodes), [&](int a, int b) { return tree.tin
            [a] < tree.tin[b]; });
        nodes.erase(unique(all(nodes)), nodes.end());
        for (int u : nodes) virt_adj[u].clear();
        vi stk(nodes[0]);
        for (int i = 1; i < nodes.size(); i++) {
            int u = nodes[i];
            while (!stk.empty() && !bl.is_ancestor(stk.back(),
                u)) stk.pop_back();
            if (!stk.empty()) {
                ll w = bl.dist_weighted(stk.back(), u);
                virt_adj[stk.back()].pb({u, w});
                virt_adj[u].pb({stk.back(), w});
            }
            stk.pb(u);
        }
        return {nodes[0], nodes};
    }
};

/**
 * Uso[1]: TreeHashing::are_isomorphic(n1, adj1, n2,
 * adj2);
 * Determina si dos arboles son isomorfos (sin
 * considerar etiquetas) mediante hashing y centroides
 * .
 * Complejidad:  $O(N)$ .
 */
struct TreeHashing {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;

```

```

        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    static uint64_t hash_node(uint64_t x) { return
        splitmix64(x + 0x517cc1b727220a95); }
    static uint64_t rooted_hash(int u, int p, const vector
        <vi>& adj) {
        vector<uint64_t> child;
        for (int v : adj[u]) if (v != p) child.pb(
            rooted_hash(v, u, adj));
        sort(all(child));
        uint64_t h = 1;
        for (uint64_t c : child) h += hash_node(c);
        return splitmix64(h);
    }
    static vector<int> get_centers(int n, const vector<vi
        >& adj) {
        vi deg(n + 1);
        queue<int> leaves;
        for (int i = 1; i <= n; i++) {
            deg[i] = adj[i].size();
            if (deg[i] <= 1) leaves.push(i);
        }
        int remaining = n;
        while (remaining > 2) {
            int sz = leaves.size();
            remaining -= sz;
            for (int i = 0; i < sz; i++) {
                int u = leaves.front(); leaves.pop();
                for (int v : adj[u]) if (--deg[v] == 1) leaves.
                    push(v);
            }
        }
        vi centers;
        while (!leaves.empty()) centers.pb(leaves.front()),
            leaves.pop();
        return centers;
    }
    static bool are_isomorphic(int n1, const vector<vi>&
        adj1, int n2, const vector<vi>& adj2) {
        if (n1 != n2) return false;
        auto c1 = get_centers(n1, adj1), c2 = get_centers(n2,
            adj2);
        for (int r1 : c1)
            for (int r2 : c2)
                if (rooted_hash(r1, 0, adj1) == rooted_hash(r2,
                    0, adj2)) return true;
        return false;
    }
};

/**
 * Uso[1]: RerootingDP r(n); r.add_edge(u, v); auto ans
 * = r.solve(root);
 * DP con rerooting para calcular la solucion para todas
 * las posibles raices; ajustar merge, extend y
 * finalize.
 * Complejidad:  $O(N)$ .
 */
struct RerootingDP {
    struct State { ll val; };
    int n;
    vector<vi> adj;

```

```

vector<State> dp_down, ans;
const State NEUTRAL = {0};
State merge(State a, State b) { return {max(a.val, b.
    val)}; }
State extend(State s, int v, int u) { return {s.val +
    1}; } // transicion por arista
State finalize(State s, int u) { return s; }
RerootingDP(int n) : n(n), adj(n + 1), dp_down(n + 1,
    NEUTRAL), ans(n + 1, NEUTRAL) {}
void add_edge(int u, int v) { adj[u].pb(v); adj[v].pb(
    u); }
void dfs_down(int u, int p) {
    State cur = NEUTRAL;
    for (int v : adj[u]) if (v != p) dfs_down(v, u), cur
        = merge(cur, extend(dp_down[v], v, u));
    dp_down[u] = finalize(cur, u);
}
void dfs_reroot(int u, int p, State from_parent) {
    vi children;
    for (int v : adj[u]) if (v != p) children.pb(v);
    int k = children.size();
    vector<State> pref(k + 1, NEUTRAL), suff(k + 1,
        NEUTRAL);
    for (int i = 0; i < k; i++) pref[i + 1] = merge(pref
        [i], extend(dp_down[children[i]], children[i], u));
    for (int i = k - 1; i >= 0; i--) suff[i] = merge(
        suff[i + 1], extend(dp_down[children[i]], children
        [i], u));
    ans[u] = finalize(merge(pref[k], from_parent), u);
    for (int i = 0; i < k; i++) {
        State outside = merge(merge(pref[i], suff[i + 1]),
            from_parent);
        dfs_reroot(children[i], u, extend(finalize(outside
            , u), u, children[i]));
    }
}
vector<State> solve(int root = 1) {
    dfs_down(root, 0);
    dfs_reroot(root, 0, NEUTRAL);
    return ans;
}
};

```

```

/**
 * Uso[1]: FastLCA<LL> fl(t); fl.lca(u, v); fl.
 * dist_edges(u, v);
 * LCA por Euler + RMQ con respuesta en tiempo constante
 * por consulta.
 * Complejidad: preproceso  $O(N \log N)$ , consulta  $O(1)$ .
 */
template <typename T = ll> struct FastLCA {
    const Tree<T>& tree;
    vi tin, euler, euler_depth, log_table;
    vector<vi> st;
    FastLCA(const Tree<T>& tree) : tree(tree), tin(tree.n
        + 1) {
        auto dfs = [&](auto&& self, int u, int p, int d) ->
            void {
                tin[u] = euler.size();
                euler.pb(u);
                euler_depth.pb(d);
                for (auto& e : tree.adj[u]) if (e.to != p) {
                    self(self, e.to, u, d + 1);
                }
            };
        euler.pb(u);
        euler_depth.pb(d);
        dfs(dfs, tree.root, 0, 0);
        int m = euler.size();
        log_table.assign(m + 1, 0);
        for (int i = 2; i <= m; i++) log_table[i] =
            log_table[i / 2] + 1;
        int K = log_table[m] + 1;
        st.assign(K, vi(m));
        for (int i = 0; i < m; i++) st[0][i] = i;
        for (int i = 1; i < K; i++)
            for (int j = 0; j + (1 << i) <= m; j++) {
                int l = st[i - 1][j], r = st[i - 1][j + (1 << i)
                    - 1]);
                st[i][j] = euler_depth[l] < euler_depth[r] ? l :
                    r;
            }
        int lca(int u, int v) {
            int l = tin[u], r = tin[v];
            if (l > r) swap(l, r);
            int i = log_table[r - l + 1];
            int a = st[i][l], b = st[i][r - (1 << i) + 1];
            return euler_depth[a] < euler_depth[b] ? euler[a] :
                euler[b];
        }
        int dist_edges(int u, int v) { return tree.depth[u] +
            tree.depth[v] - 2 * tree.depth[lca(u, v)]; }
    };

```

```

    euler.pb(u);
    euler_depth.pb(d);
    };
    dfs(dfs, tree.root, 0, 0);
    int m = euler.size();
    log_table.assign(m + 1, 0);
    for (int i = 2; i <= m; i++) log_table[i] =
        log_table[i / 2] + 1;
    int K = log_table[m] + 1;
    st.assign(K, vi(m));
    for (int i = 0; i < m; i++) st[0][i] = i;
    for (int i = 1; i < K; i++)
        for (int j = 0; j + (1 << i) <= m; j++) {
            int l = st[i - 1][j], r = st[i - 1][j + (1 << i)
                - 1]);
            st[i][j] = euler_depth[l] < euler_depth[r] ? l :
                r;
        }
    int lca(int u, int v) {
        int l = tin[u], r = tin[v];
        if (l > r) swap(l, r);
        int i = log_table[r - l + 1];
        int a = st[i][l], b = st[i][r - (1 << i) + 1];
        return euler_depth[a] < euler_depth[b] ? euler[a] :
            euler[b];
    }
    int dist_edges(int u, int v) { return tree.depth[u] +
        tree.depth[v] - 2 * tree.depth[lca(u, v)]; }
};
/**
 * Uso[1]: MoOnTrees<LL> mo(n); mo.dfs_euler(root, 0, t)
 * ; mo.solve(queries, fast_lca);
 * Algoritmo de Mo sobre caminos de un arbol; ajustar
 * toggle para actualizar la respuesta al agregar o
 * quitar un nodo.
 * Complejidad:  $O((N + Q)\sqrt{N})$  toggle esperados.
 */
template <typename T = ll> struct MoOnTrees {
    struct Query {
        int l, r, lca, id, block_size;
        bool operator<(const Query& o) const {
            int b1 = l / block_size, b2 = o.l / block_size;
            if (b1 != b2) return b1 < b2;
            return (b1 & 1) ? (r < o.r) : (r > o.r);
        }
    };
    int n, timer;
    vi tin, tout, euler;
    vector<bool> in_path;
    ll current_ans = 0;
    MoOnTrees(int n) : n(n), timer(0), tin(n + 1), tout(n
        + 1), euler(2 * n + 1), in_path(n + 1, false) {}
    void dfs_euler(int u, int p, const Tree<T>& tree) {
        tin[u] = ++timer;
        euler[timer] = u;
        for (auto& e : tree.adj[u]) if (e.to != p) dfs_euler
            (e.to, u, tree);
        tout[u] = ++timer;
        euler[timer] = u;
    }
};

```

```

void toggle(int u) {
    // TODO: actualizar respuesta al agregar/remover el
    // nodo u (p.ej. frecuencias de colores)
    in_path[u] = !in_path[u];
}
vll solve(vector<pair<int, int>>& raw_queries, const
    FastLCA<T>& fast_lca) {
    int q = raw_queries.size();
    int bs = max(1, (int)(2 * n / sqrt(max(1, q))));
    vector<Query> queries;
    for (int i = 0; i < q; i++) {
        auto [u, v] = raw_queries[i];
        if (tin[u] > tin[v]) swap(u, v);
        int anc = fast_lca.lca(u, v);
        queries.pb(anc == u ? Query{tin[u], tin[v], 0, i,
            bs} : Query{tout[u], tin[v], anc, i, bs});
    }
    sort(all(queries));
    vll ans(q);
    int curr_l = 1, curr_r = 0;
    for (auto& mq : queries) {
        while (curr_l > mq.l) toggle(euler[--curr_l]);
        while (curr_r < mq.r) toggle(euler[++curr_r]);
        while (curr_l < mq.l) toggle(euler[curr_l++]);
        while (curr_r > mq.r) toggle(euler[curr_r--]);
        if (mq.lca) toggle(mq.lca);
        ans[mq.id] = current_ans;
        if (mq.lca) toggle(mq.lca);
    }
    return ans;
}
};

```

Range Queries

```

/**
 * Uso[0]: SparseTable st(a); st.query(L, R);
 * Consultas estaticas en rangos para operaciones
 * idempotentes (min, max, gcd).
 * Complejidad: construccion  $O(N \log N)$ , consulta  $O(1)$ .
 */
struct SparseTable {
    int n;
    vector<vll> st;
    SparseTable(const vll& a) : n(a.size()) {
        int log_n = log2(n) + 1;
        st.assign(n, vll(log_n));
        for (int i = 0; i < n; i++) st[i][0] = a[i];
        for (int j = 1; j < log_n; j++)
            for (int i = 0; i + (1 << j) <= n; i++)
                st[i][j] = min(st[i][j - 1], st[i + (1 << j) -
                    1][j - 1]); // min/max/gcd
    }
    ll query(int L, int R) {
        int j = log2(R - L + 1);
        return min(st[L][j], st[R - (1 << j) + 1][j]);
    }
};
/**
 * Uso[1]: FenwickTree ft(n); ft.add(i, x); ll s = ft.
 * query(2, 3);
 */

```

```

* Arbol de Fenwick para sumas de prefijos y rangos con
  actualizaciones puntuales.
* Complejidad: actualizacion y consulta  $O(\log N)$ .
*/
struct FenwickTree {
    int n;
    vll bit;
    FenwickTree(int n) : n(n), bit(n + 1, 0) {}
    void add(int idx, ll val) { for (; idx <= n; idx +=
        idx & -idx) bit[idx] += val; }
    ll sum(int idx) {
        ll res = 0;
        for (; idx > 0; idx -= idx & -idx) res += bit[idx];
        return res;
    }
    ll query(int L, int R) { return sum(R) - sum(L - 1); }
};

/**
* Uso[0]: IterativeSegTree st(a); st.update(pos, val);
  st.query(L, R);
* Arbol de segmentos iterativo para suma (o min/max)
  con actualizacion puntual.
* Complejidad: actualizacion y consulta  $O(\log N)$ .
*/
struct IterativeSegTree {
    int n;
    vll st;
    IterativeSegTree(const vll& a) : n(a.size()), st(2 * a
        .size(), 0) {
        for (int i = 0; i < n; i++) st[n + i] = a[i];
        for (int i = n - 1; i > 0; i--) st[i] = st[i < 1] +
            st[i < 1 | 1]; // + / min / max
    }
    void update(int pos, ll val) {
        for (st[pos += n] = val; pos > 1; pos >>= 1) st[pos
            >> 1] = st[pos] + st[pos ^ 1];
    }
    ll query(int L, int R) {
        ll resL = 0, resR = 0;
        for (L += n, R += n + 1; L < R; L >>= 1, R >>= 1) {
            if (L & 1) resL += st[L++];
            if (R & 1) resR += st[--R];
        }
        return resL + resR;
    }
};

/**
* Uso[0]: RecursiveSegTree st(a); st.update(pos, val);
  st.query(L, R); st.find_first(val);
* Arbol de segmentos recursivo para max (o suma/min)
  con busqueda del primer indice con valor mayor o
  igual.
* Complejidad: actualizacion y consulta  $O(\log N)$ .
*/
struct RecursiveSegTree {
    int n;
    vll st;
    ll combine(ll a, ll b) { return max(a, b); } // max /
        + / min
    RecursiveSegTree(const vll& a) : n(a.size()), st(4 * a
        .size()) { build(1, 0, n - 1, a); }

```

```

    void build(int p, int L, int R, const vll& a) {
        if (L == R) { st[p] = a[L]; return; }
        int mid = (L + R) / 2;
        build(p << 1, L, mid, a);
        build(p << 1 | 1, mid + 1, R, a);
        st[p] = combine(st[p << 1], st[p << 1 | 1]);
    }
    void update(int p, int L, int R, int pos, ll val) {
        if (L == R) { st[p] = val; return; } // o st[p] +=
            val
        int mid = (L + R) / 2;
        pos <= mid ? update(p << 1, L, mid, pos, val) :
            update(p << 1 | 1, mid + 1, R, pos, val);
        st[p] = combine(st[p << 1], st[p << 1 | 1]);
    }
    ll query(int p, int L, int R, int qL, int qR) {
        if (qL > R || qR < L) return 0; // NEUTRAL
        if (qL <= L && R <= qR) return st[p];
        int mid = (L + R) / 2;
        return combine(query(p << 1, L, mid, qL, qR), query(
            p << 1 | 1, mid + 1, R, qL, qR));
    }
    // Primer indice con valor >= val (asume arbol de max)
    // -1 si no existe
    int find_first(int p, int L, int R, ll val) {
        if (st[p] < val) return -1;
        if (L == R) return L;
        int mid = (L + R) / 2;
        return st[p << 1] >= val ? find_first(p << 1, L, mid
            , val) : find_first(p << 1 | 1, mid + 1, R, val);
    }
    void update(int pos, ll val) { update(1, 0, n - 1, pos
        , val); }
    ll query(int L, int R) { return query(1, 0, n - 1, L,
        R); }
    int find_first(ll val) { return find_first(1, 0, n -
        1, val); }
};

/**
* Uso[0]: LazySegTree st(a); st.add_range(L, R, v); st.
  set_range(L, R, v); st.query(L, R); st.get(pos);
* Arbol de segmentos con propagacion perezosa para suma
  con actualizaciones de rango (add y set).
* Complejidad: actualizacion y consulta de rango
   $O(\log N)$ .
*/
struct LazySegTree {
    int n;
    vll st, lazy_add, lazy_set;
    vector<bool> marked;
    LazySegTree(const vll& a) : n(a.size()), st(4 * a.size()
        ()), lazy_add(4 * a.size()),
        lazy_set(4 * a.size(), false) {
        build(1, 0, n - 1, a); }
    void apply_set(int p, ll val, int L, int R) {
        st[p] = val * (R - L + 1); // para min/max: st[p] =
            val
        lazy_set[p] = val;
        lazy_add[p] = 0;
        marked[p] = true;
    }

```

```

    void apply_add(int p, ll val, int L, int R) {
        st[p] += val * (R - L + 1); // para min/max: st[p]
            += val
        if (marked[p]) lazy_set[p] += val;
        else lazy_add[p] += val;
    }
    void push(int p, int L, int R) {
        int mid = (L + R) / 2;
        if (marked[p]) {
            apply_set(p << 1, lazy_set[p], L, mid);
            apply_set(p << 1 | 1, lazy_set[p], mid + 1, R);
            marked[p] = false;
        }
        if (lazy_add[p]) {
            apply_add(p << 1, lazy_add[p], L, mid);
            apply_add(p << 1 | 1, lazy_add[p], mid + 1, R);
            lazy_add[p] = 0;
        }
    }
    void build(int p, int L, int R, const vll& a) {
        if (L == R) { st[p] = a[L]; return; }
        int mid = (L + R) / 2;
        build(p << 1, L, mid, a);
        build(p << 1 | 1, mid + 1, R, a);
        st[p] = st[p << 1] + st[p << 1 | 1];
    }
    void update(int p, int L, int R, int qL, int qR, ll
        val, int type) { // 1 = add, 2 = set
        if (qL > R || qR < L) return;
        if (qL <= L && R <= qR) { type == 1 ? apply_add(p,
            val, L, R) : apply_set(p, val, L, R); return; }
        push(p, L, R);
        int mid = (L + R) / 2;
        update(p << 1, L, mid, qL, qR, val, type);
        update(p << 1 | 1, mid + 1, R, qL, qR, val, type);
        st[p] = st[p << 1] + st[p << 1 | 1];
    }
    ll query(int p, int L, int R, int qL, int qR) {
        if (qL > R || qR < L) return 0;
        if (qL <= L && R <= qR) return st[p];
        push(p, L, R);
        int mid = (L + R) / 2;
        return query(p << 1, L, mid, qL, qR) + query(p << 1
            | 1, mid + 1, R, qL, qR);
    }
    ll get_point(int p, int L, int R, int pos) {
        if (L == R) return st[p];
        push(p, L, R);
        int mid = (L + R) / 2;
        return pos <= mid ? get_point(p << 1, L, mid, pos) :
            get_point(p << 1 | 1, mid + 1, R, pos);
    }
    void add_range(int L, int R, ll val) { update(1, 0, n
        - 1, L, R, val, 1); }
    void set_range(int L, int R, ll val) { update(1, 0, n
        - 1, L, R, val, 2); }
    ll query(int L, int R) { return query(1, 0, n - 1, L,
        R); }
    ll get(int pos) { return get_point(1, 0, n - 1, pos);
    }
};

```

```

/**
 * Uso[1]: DynamicSegTree st(max_rango); st.update(pos,
 * val); st.query(L, R);
 * Arbol de segmentos dinamico (sparse) para indices
 * grandes sin compresion previa.
 * Complejidad: actualizacion y consulta  $O(\log \text{max\_rango})$ 
 */
struct DynamicSegTree {
    struct Node { ll sum = 0; int left = 0, right = 0; };
    vector<Node> st;
    ll max_range;
    int root;
    DynamicSegTree(ll max_range = 1e9) : max_range(
        max_range), root(0) { st.emplace_back(); }
    int new_node() { st.emplace_back(); return st.size() -
        1; }
    int update(int node, ll L, ll R, ll pos, ll val) {
        if (!node) node = new_node();
        st[node].sum += val;
        if (L == R) return node;
        ll mid = (L + R) / 2;
        if (pos <= mid) st[node].left = update(st[node].left
            , L, mid, pos, val);
        else st[node].right = update(st[node].right, mid +
            1, R, pos, val);
        return node;
    }
    ll query(int node, ll L, ll R, ll qL, ll qR) {
        if (!node || qL > R || qR < L) return 0;
        if (qL <= L && R <= qR) return st[node].sum;
        ll mid = (L + R) / 2;
        return query(st[node].left, L, mid, qL, qR) + query(
            st[node].right, mid + 1, R, qL, qR);
    }
    void update(ll pos, ll val) { root = update(root, 1,
        max_range, pos, val); }
    ll query(ll L, ll R) { return query(root, 1, max_range
        , L, R); }
};

/**
 * Uso[0]: PersistentSegTree pst(a); int v = pst.update(
 * version, pos, val); pst.query(v, L, R);
 * Arbol de segmentos persistente con versiones
 * historicas y consultas sobre cualquier version.
 * Complejidad: actualizacion y consulta  $O(\log N)$ ;
 * espacio  $O(N \log N)$ .
 */
struct PersistentSegTree {
    struct Node { ll sum; int left, right; Node(ll s = 0,
        int l = 0, int r = 0) : sum(s), left(l), right(r)
        {} };
    vector<Node> st;
    vi roots;
    int n;
    int new_node(ll sum, int left, int right) { st.
        emplace_back(sum, left, right); return st.size() -
        1; }
    int build(int L, int R, const vll& a) {
        if (L == R) return new_node(a[L], 0, 0);
        int mid = (L + R) / 2;
        int l = build(L, mid, a), r = build(mid + 1, R, a);

```

```

        return new_node(st[l].sum + st[r].sum, l, r);
    }
    int update(int old, int L, int R, int pos, ll val) {
        if (L == R) return new_node(st[old].sum + val, 0, 0)
            ;
        int mid = (L + R) / 2;
        int l = st[old].left, r = st[old].right;
        if (pos <= mid) l = update(l, L, mid, pos, val);
        else r = update(r, mid + 1, R, pos, val);
        return new_node(st[l].sum + st[r].sum, l, r);
    }
    ll query(int node, int L, int R, int qL, int qR) {
        if (!node || qL > R || qR < L) return 0;
        if (qL <= L && R <= qR) return st[node].sum;
        int mid = (L + R) / 2;
        return query(st[node].left, L, mid, qL, qR) + query(
            st[node].right, mid + 1, R, qL, qR);
    }
    PersistentSegTree(const vll& a) : n(a.size()) {
        st.emplace_back();
        roots.pb(build(0, n - 1, a));
    }
    int update(int prev_version, int pos, ll val) { //
        crea nueva version
        roots.pb(update(roots[prev_version], 0, n - 1, pos,
            val));
        return roots.size() - 1;
    }
    ll query(int version, int L, int R) { return query(
        roots[version], 0, n - 1, L, R); }
};

/**
 * Uso[1]: FenwickTree2D ft(n, m); ft.add(r, c, v); ft.
 * query(r1, c1, r2, c2);
 * Arbol de Fenwick bidimensional para sumas de
 * submatrices con actualizaciones puntuales.
 * Complejidad: actualizacion y consulta  $O(\log n \cdot \log m)$ .
 */
struct FenwickTree2D {
    int n, m;
    vector<vll> bit;
    FenwickTree2D(int n, int m) : n(n), m(m), bit(n + 1,
        vll(m + 1, 0)) {}
    void add(int r, int c, ll val) {
        for (int i = r; i <= n; i += i & -i)
            for (int j = c; j <= m; j += j & -j) bit[i][j] +=
                val;
    }
    ll sum(int r, int c) {
        ll res = 0;
        for (int i = r; i > 0; i -= i & -i)
            for (int j = c; j > 0; j -= j & -j) res += bit[i][
                j];
        return res;
    }
    ll query(int r1, int c1, int r2, int c2) {
        return sum(r2, c2) - sum(r1 - 1, c2) - sum(r2, c1 -
            1) + sum(r1 - 1, c1 - 1);
    }
};

/**

```

```

 * Uso[0]: MergeSortTree st(a); st.count_less_than(L, R,
 * val); st.count_less_equal(L, R, val);
 * Arbol de segmentos con listas ordenadas para contar
 * elementos menores o iguales a un valor en un rango.
 * Complejidad: construccion  $O(N \log N)$ , consulta
 *  $O(\log^2 N)$ .
 */
struct MergeSortTree {
    int n;
    vector<vll> st;
    MergeSortTree(const vll& a) : n(a.size()), st(4 * a.
        size()) { build(1, 0, n - 1, a); }
    void build(int p, int L, int R, const vll& a) {
        if (L == R) { st[p] = {a[L]}; return; }
        int mid = (L + R) / 2;
        build(p << 1, L, mid, a);
        build(p << 1 | 1, mid + 1, R, a);
        st[p].resize(st[p << 1].size() + st[p << 1 | 1].size
            ());
        merge(all(st[p << 1]), all(st[p << 1 | 1]), st[p].
            begin());
    }
    int less_than(int p, int L, int R, int qL, int qR, ll
        val) {
        if (qL > R || qR < L) return 0;
        if (qL <= L && R <= qR) return lower_bound(all(st[p
            ]), val) - st[p].begin();
        int mid = (L + R) / 2;
        return less_than(p << 1, L, mid, qL, qR, val) +
            less_than(p << 1 | 1, mid + 1, R, qL, qR, val);
    }
    int less_equal(int p, int L, int R, int qL, int qR, ll
        val) {
        if (qL > R || qR < L) return 0;
        if (qL <= L && R <= qR) return upper_bound(all(st[p
            ]), val) - st[p].begin();
        int mid = (L + R) / 2;
        return less_equal(p << 1, L, mid, qL, qR, val) +
            less_equal(p << 1 | 1, mid + 1, R, qL, qR, val);
    }
    int count_less_than(int L, int R, ll val) { return
        less_than(1, 0, n - 1, L, R, val); }
    int count_less_equal(int L, int R, ll val) { return
        less_equal(1, 0, n - 1, L, R, val); }
};

/**
 * Uso[0]: SqrtDecomposition sq(a); sq.update(pos, val);
 * sq.query(L, R);
 * Descomposicion en bloques para sumas de rangos con
 * actualizaciones puntuales.
 * Complejidad: actualizacion y consulta  $O(\sqrt{N})$ .
 */
struct SqrtDecomposition {
    int n, block_size;
    vll a, blocks;
    SqrtDecomposition(const vll& arr) : n(arr.size()),
        block_size(max(1, (int)sqrt(n))), a(arr) {
        blocks.assign((n + block_size - 1) / block_size, 0);
        for (int i = 0; i < n; i++) blocks[i / block_size]
            += a[i];
    }

```

```

void update(int pos, ll val) {
    int b = pos / block_size;
    a[pos] = val;
    blocks[b] = 0;
    int end = min(n, (b + 1) * block_size);
    for (int i = b * block_size; i < end; i++) blocks[b]
        += a[i];
}
ll query(int L, int R) {
    int bL = L / block_size, bR = R / block_size;
    ll res = 0;
    if (bL == bR) {
        for (int i = L; i <= R; i++) res += a[i];
        return res;
    }
    for (int i = L; i < (bL + 1) * block_size; i++) res
        += a[i];
    for (int b = bL + 1; b < bR; b++) res += blocks[b];
    for (int i = bR * block_size; i <= R; i++) res += a[
        i];
    return res;
}
};

mt19937_64 rng(chrono::steady_clock::now().
    time_since_epoch().count());
struct TreapNode {
    ll key, prior;
    int sz;
    TreapNode *l, *r;
    TreapNode(ll key) : key(key), prior(rng()), sz(1), l(
        nullptr), r(nullptr) {}
};
typedef TreapNode* pNode;
/**
 * Uso[0]: Treap t; t.insert(x); t.erase(x); t.
    kth_element(k); t.count_less_than(x);
 * Arbol binario de busqueda balanceado aleatoriamente (
    treap) con consultas por orden.
 * Complejidad: cada operacion O(log N) esperado.
 */
struct Treap {
    pNode root = nullptr;
    int sz(pNode t) { return t ? t->sz : 0; }
    void upd(pNode t) { if (t) t->sz = 1 + sz(t->l) + sz(t
        ->r); }
    void split(pNode t, ll key, pNode& l, pNode& r) { // L
        : keys <= key
        if (!t) { l = r = nullptr; return; }
        if (key >= t->key) split(t->r, key, t->r, r), l = t;
        else split(t->l, key, l, t->l), r = t;
        upd(t);
    }
    void merge(pNode& t, pNode l, pNode r) {
        if (!l || !r) { t = l ? l : r; return; }
        if (l->prior > r->prior) merge(l->r, l->r, r), t = l
            ;
        else merge(r->l, l, r->l), t = r;
        upd(t);
    }
    void insert(ll key) {
        pNode l, r;
        split(root, key, l, r);
        pNode node = new TreapNode(key);
        merge(l, l, node);
        merge(root, l, r);
    }
    void erase(ll key) {
        pNode l, m, r;
        split(root, key - 1, l, r);
        split(r, key, m, r);
        if (m) {
            pNode tmp = m;
            merge(m, m->l, m->r);
            delete tmp;
        }
        merge(root, l, m);
        merge(root, root, r);
    }
    ll kth_element(int k) { // k-esimo menor (0-indexed).
        LINf si fuera de rango
        pNode cur = root;
        while (cur) {
            int ls = sz(cur->l);
            if (ls == k) return cur->key;
            if (ls > k) cur = cur->l;
            else k -= ls + 1, cur = cur->r;
        }
        return LINf;
    }
    int count_less_than(ll key) {
        pNode l, r;
        split(root, key - 1, l, r);
        int res = sz(l);
        merge(root, l, r);
        return res;
    }
};
/**
 * Uso[0]: ImplicitTreap it; it.insert(pos, val); it.
    erase(pos); it.query_range(L, R); it.reverse_range(
    L, R);
 * Treap implicito para secuencias dinamicas con
    insercion/borrado por posicion, suma de rangos y
    reversiones perezosas.
 * Complejidad: cada operacion O(log N) esperado.
 */
struct ImplicitNode {
    ll val, sum, prior;
    int sz;
    bool lazy_rev;
    ImplicitNode *l, *r;
    ImplicitNode(ll val) : val(val), sum(val), prior(rng(
        )), sz(1), lazy_rev(false), l(nullptr), r(nullptr)
        {}
};
typedef ImplicitNode* pINode;
struct ImplicitTreap {
    pINode root = nullptr;
    int sz(pINode t) { return t ? t->sz : 0; }
    ll sum(pINode t) { return t ? t->sum : 0; }
    void push(pINode t) {
        if (t && t->lazy_rev) {
            t->lazy_rev = false;
            swap(t->l, t->r);
        }
        if (t->l) t->l->lazy_rev ^= true;
        if (t->r) t->r->lazy_rev ^= true;
    }
    void upd(pINode t) {
        if (t) t->sz = 1 + sz(t->l) + sz(t->r), t->sum = t->
            val + sum(t->l) + sum(t->r);
    }
    void split(pINode t, int k, pINode& l, pINode& r) { //
        L: primeros k elementos
        if (!t) { l = r = nullptr; return; }
        push(t);
        int implicit = sz(t->l) + 1;
        if (k >= implicit) split(t->r, k - implicit, t->r, r
            ), l = t;
        else split(t->l, k, l, t->l), r = t;
        upd(t);
    }
    void merge(pINode& t, pINode l, pINode r) {
        push(l); push(r);
        if (!l || !r) { t = l ? l : r; return; }
        if (l->prior > r->prior) merge(l->r, l->r, r), t = l
            ;
        else merge(r->l, l, r->l), t = r;
        upd(t);
    }
    void insert(int pos, ll val) {
        pINode l, r;
        split(root, pos, l, r);
        pINode node = new ImplicitNode(val);
        merge(l, l, node);
        merge(root, l, r);
    }
    void erase(int pos) {
        pINode l, mid, r;
        split(root, pos, l, r);
        split(r, 1, mid, r);
        delete mid;
        merge(root, l, r);
    }
    void reverse_range(int L, int R) {
        pINode l, mid, r;
        split(root, L, l, mid);
        split(mid, R - L + 1, mid, r);
        mid->lazy_rev ^= true;
        merge(root, l, mid);
        merge(root, root, r);
    }
    ll query_range(int L, int R) {
        pINode l, mid, r;
        split(root, L, l, mid);
        split(mid, R - L + 1, mid, r);
        ll ans = sum(mid);
        merge(root, l, mid);
        merge(root, root, r);
        return ans;
    }
};
/**
 * Uso[0]: Mo mo(n); mo.add_query(L, r); auto ans = mo.
    solve(add, remove, get_ans);
 * Algoritmo de Mo 1D para responder consultas de rango

```

```

    offline cuando add y remove son  $O(1)$ .
    * Complejidad:  $O((N + Q)\sqrt{N})$  movimientos.
    */
struct Mo {
    struct Query { int l, r, id, block; };
    int n, bs;
    vector<Query> qs;
    Mo(int n, int bs = 0) : n(n), bs(bs ? bs : max(1, (int)
        sqrt(n))) {}
    void add_query(int l, int r) { qs.pb({l, r, (int)qs.
        size(), 1 / bs}); }
    template <class Add, class Remove, class GetAns>
    vector<ll> solve(Add add, Remove remove, GetAns
        get_ans) {
        vector<ll> res(qs.size());
        sort(all(qs), [](const Query& a, const Query& b) {
            if (a.block != b.block) return a.block < b.block;
            return (a.block & 1) ? a.r > b.r : a.r < b.r; //
                odd-even para acelerar
        });
        int cur_l = 0, cur_r = -1;
        for (const Query& q : qs) {
            while (cur_l > q.l) add(--cur_l);
            while (cur_r < q.r) add(++cur_r);
            while (cur_l < q.l) remove(cur_l++);
            while (cur_r > q.r) remove(cur_r--);
            res[q.id] = get_ans();
        }
        return res;
    }
};

/**
    * Uso[0]: MoWithUpdates mo(n); mo.add_update(pos, antes
        , despues); mo.add_query(l, r); auto ans = mo.solve
        (add, remove, apply, revert, get_ans);
    * Algoritmo de Mo con actualizaciones puntuales (Mo 3D)
        ; responde consultas sobre el arreglo en un
        instante dado.
    * Complejidad:  $O((N + Q)^{5/3})$  movimientos esperados.
    */
struct MoWithUpdates {
    struct Query { int l, r, t, id; };
    struct Update { int pos; ll old_val, new_val; };
    int n, bs;
    vector<Query> qs;
    vector<Update> ups;
    MoWithUpdates(int n, int bs = 0) : n(n), bs(bs ? bs :
        max(1, (int)pow(n, 2.0 / 3.0))) {}
    void add_update(int pos, ll old_val, ll new_val) { ups
        .pb({pos, old_val, new_val}); }
    void add_query(int l, int r) { qs.pb({l, r, (int)ups.
        size(), (int)qs.size()}); }
    template <class Add, class Remove, class Apply, class
        Revert, class GetAns>
    vector<ll> solve(Add add, Remove remove, Apply apply,
        Revert revert, GetAns get_ans) {
        vector<ll> res(qs.size());
        sort(all(qs), [](const Query& a, const Query& b) {
            int ba = a.l / bs, bb = b.l / bs;
            if (ba != bb) return ba < bb;
            int ca = a.r / bs, cb = b.r / bs;
            if (ca != cb) return ca < cb;

```

```

            return a.t < b.t;
        });
        int cur_l = 0, cur_r = -1, cur_t = 0;
        for (const Query& q : qs) {
            while (cur_t < q.t) { apply(ups[cur_t], cur_l,
                cur_r); cur_t++; }
            while (cur_t > q.t) { cur_t--; revert(ups[cur_t],
                cur_l, cur_r); }
            while (cur_l > q.l) add(--cur_l);
            while (cur_r < q.r) add(++cur_r);
            while (cur_l < q.l) remove(cur_l++);
            while (cur_r > q.r) remove(cur_r--);
            res[q.id] = get_ans();
        }
        return res;
    }
};

/**
    * Uso[0]: RollbackMo mo(n); mo.add_query(l, r); mo.
        solve(add, current, snapshot, rollback, reset);
    * Variante de Mo con rollback para estructuras donde
        agregar es  $O(1)$  pero eliminar es costoso o
        imposible.
    * Complejidad:  $O((N + Q)\sqrt{N})$  operaciones de add.
    */
struct RollbackMo {
    struct Query { int l, r, id; };
    int n, bs;
    vector<Query> qs;
    RollbackMo(int n, int bs = 0) : n(n), bs(bs ? bs : max
        (1, (int)sqrt(n))) {}
    void add_query(int l, int r) { qs.pb({l, r, (int)qs.
        size()}); }
    template <class Add, class Current, class Snapshot,
        class Rollback, class Reset>
    vector<ll> solve(Add add, Current current, Snapshot
        snapshot, Rollback rollback, Reset reset) {
        vector<ll> res(qs.size());
        sort(all(qs), [](const Query& a, const Query& b) {
            if (a.l / bs != b.l / bs) return a.l / bs < b.l /
                bs;
            return a.r < b.r;
        });
        int block = -1, cur_r = -1;
        for (const Query& q : qs) {
            int b = q.l / bs;
            int block_end = min(n - 1, (b + 1) * bs - 1);
            if (b != block) {
                block = b;
                cur_r = block_end; // rango persistente vacio
                reset();
            }
            if (q.r <= block_end) {
                // query contenida en el bloque: brute force
                sobre estructura vacia
                snapshot();
                for (int i = q.l; i <= q.r; i++) add(i);
                res[q.id] = current();
                rollback();
            } else {
                while (cur_r < q.r) add(++cur_r);
                snapshot();

```

```

                for (int i = block_end; i >= q.l; i--) add(i);
                res[q.id] = current();
                rollback();
            }
        }
        return res;
    }
};

```

Strings

```

/**
    * Uso: to_lower(s), to_upper(s), trim(s), replace_all(s
        , "a", "b"), split(s, ','), join(v, " ");
    * Utilidades de procesamiento de strings: mayusculas/
        minusculas, recorte, reemplazo, division y union.
    * Complejidad:  $O(N)$  por operacion.
    */
void to_lower(string& s) { transform(all(s), s.begin(),
    ::tolower); }
void to_upper(string& s) { transform(all(s), s.begin(),
    ::toupper); }
void trim(string& s) {
    s.erase(s.begin(), find_if(all(s), [](unsigned char c)
        { return !isspace(c); }));
    s.erase(find_if(rall(s), [](unsigned char c) { return
        !isspace(c); }).base(), s.end());
}
void replace_all(string& s, const string& old, const
    string& nw) {
    size_t pos = 0;
    while ((pos = s.find(old, pos)) != string::npos) s.
        replace(pos, old.size(), nw, pos += nw.size());
}
vector<string> split(const string& s, char delim) {
    vector<string> res;
    string item;
    istringstream iss(s);
    while (getline(iss, item, delim)) res.pb(item);
    return res;
}
string join(const vector<string>& v, const string& delim
    ) {
    string res;
    for (int i = 0; i < v.size(); i++) {
        res += v[i];
        if (i + 1 < v.size()) res += delim;
    }
    return res;
}

/**
    * Uso: auto pi = prefix_function(s);
    * Funcion prefijo de KMP:  $\pi[i]$  es el prefijo propio mas
        largo que tambien es sufijo de  $s[0..i]$ .
    * Complejidad:  $O(N)$ .
    */
vi prefix_function(const string& s) {
    int n = s.size();
    vi pi(n, 0);
    for (int i = 1; i < n; i++) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j]) j = pi[j - 1];

```

```

    if (s[i] == s[j]) j++;
    pi[i] = j;
}
return pi;
}

/**
 * Uso: auto z = z_function(s);
 * Funcion Z:  $z[i]$  es la longitud del prefijo de s que coincide con el sufijo que inicia en i.
 * Complejidad:  $O(N)$ .
 */
vi z_function(const string& s) {
    int n = s.size(), l = 0, r = 0;
    vi z(n, 0);
    for (int i = 1; i < n; i++) {
        if (i <= r) z[i] = min(r - i + 1, z[i - 1]);
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
        if (i + z[i] - 1 > r) l = i, r = i + z[i] - 1;
    }
    return z;
}

/**
 * Uso: string res = remove_all_occurrences(s, p);
 * Elimina todas las apariciones de un patron p en s, incluidas las encadenadas.
 * Complejidad:  $O(N + M)$  con  $N = |s|$  y  $M = |p|$ .
 */
string remove_all_occurrences(const string& s, const string& p) {
    vi pi = prefix_function(p);
    string res;
    vi state(s.size() + 1, 0);
    for (char c : s) {
        res += c;
        int j = state[res.size() - 1];
        while (j > 0 && p[j] != c) j = pi[j - 1];
        if (p[j] == c) j++;
        state[res.size()] = j;
        if (j == p.size()) res.resize(res.size() - p.size());
    }
    return res;
}

/**
 * Uso[0]: auto occ = rabin_karp(text, pattern);
 * Encuentra las posiciones donde comienza un patron usando hashing (rolling hash).
 * Complejidad:  $O(T + S)$  con  $T = |text|$  y  $S = |pattern|$ .
 */
vi rabin_karp(const string& text, const string& pattern) {
    const ll P = 31, M = 1e9 + 9;
    int S = pattern.size(), T = text.size();
    if (!S || S > T) return {};
    vll p_pow(max(S, T));
    p_pow[0] = 1;
    for (int i = 1; i < p_pow.size(); i++) p_pow[i] = p_pow[i - 1] * P % M;
    vll h(T + 1, 0);

```

```

    for (int i = 0; i < T; i++) h[i + 1] = (h[i] + (text[i] - 'a' + 1) * p_pow[i]) % M;
    ll hs = 0;
    for (int i = 0; i < S; i++) hs = (hs + (pattern[i] - 'a' + 1) * p_pow[i]) % M;
    vi occ;
    for (int i = 0; i + S - 1 < T; i++) {
        ll cur = (h[i + S] + M - h[i]) % M;
        if (cur == hs * p_pow[i] % M) occ.pb(i);
    }
    return occ;
}

/**
 * Uso: auto p = manacher(s);
 * Radios de los palindromos en cada posicion (incluyendo centros pares); el maximo da el palindromo mas largo.
 * Complejidad:  $O(N)$ .
 */
vi manacher(const string& s) {
    string t = "^#";
    for (char c : s) t += c, t += '#';
    t += '$';
    int n = t.size(), c = 0, r = 0;
    vi p(n, 0);
    for (int i = 1; i < n - 1; i++) {
        int mir = 2 * c - i;
        p[i] = r > i ? min(r - i, p[mir]) : 0;
        while (t[i + 1 + p[i]] == t[i - 1 - p[i]]) p[i]++;
        if (i + p[i] > r) c = i, r = i + p[i];
    }
    return vi(p.begin() + 1, p.end() - 1);
}

/**
 * Uso[0]: StringHash sh(s); sh.get_hash(l, r);
 * Hash doble de substrings para compararlos en  $O(1)$ .
 * Complejidad: preproceso  $O(N)$ , consulta  $O(1)$ .
 */
struct StringHash {
    static constexpr ll MOD1 = 1000000007LL; // 1e9 + 7
    static constexpr ll MOD2 = 1000000009LL; // 1e9 + 9
    static constexpr ll BASE1 = 31;
    static constexpr ll BASE2 = 37;
    string s;
    int n;
    vll h1, h2, p1, p2;
    StringHash(const string& s) : s(s), n(s.size()), h1(n + 1, 0), h2(n + 1, 0), p1(n + 1, 1), p2(n + 1, 1) {
        for (int i = 0; i < n; i++) {
            h1[i + 1] = (h1[i] * BASE1 + (s[i] - 'a' + 1)) % MOD1;
            h2[i + 1] = (h2[i] * BASE2 + (s[i] - 'a' + 1)) % MOD2;
            p1[i + 1] = p1[i] * BASE1 % MOD1;
            p2[i + 1] = p2[i] * BASE2 % MOD2;
        }
    }
    uint64_t get_hash(int l, int r) {
        ll x1 = (h1[r + 1] - h1[l] * p1[r - l + 1] % MOD1 + MOD1) % MOD1;
        ll x2 = (h2[r + 1] - h2[l] * p2[r - l + 1] % MOD2 +

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```

        MOD2) % MOD2;
        return (uint64_t)x1 << 32 | (uint32_t)x2;
    }
};

/**
 * Uso: Trie t; t.insert(s); t.search(s);
 * Trie (arbol de prefijos) sobre el alfabeto de minusculas con insercion y busqueda de palabras.
 * Complejidad: insercion y busqueda  $O(L)$  con L la longitud de la cadena.
 */
struct Trie {
    struct Node { int next[26] = {}; bool is_end = false; int count = 0; };
    vector<Node> t;
    Trie() { t.emplace_back(); }
    void insert(const string& s) {
        int v = 0;
        for (char ch : s) {
            int c = ch - 'a';
            if (!t[v].next[c]) t[v].next[c] = t.size(), t.emplace_back();
            v = t[v].next[c];
            t[v].count++;
        }
        t[v].is_end = true;
    }
    bool search(const string& s) {
        int v = 0;
        for (char ch : s) {
            int c = ch - 'a';
            if (!t[v].next[c]) return false;
            v = t[v].next[c];
        }
        return t[v].is_end;
    }
};

/**
 * Uso[0]: AhoCorasick ac; ac.insert(p); ac.build(); ac.count_occurrences(text); ac.first_occurrences(text);
 * Automata Aho-Corasick para buscar multiples patrones en un texto: conteo y primera aparicion de cada uno.
 * Complejidad: construccion  $O(\sum |p_i| \cdot 26)$ , procesamiento del texto  $O(T)$ .
 */
struct AhoCorasick {
    struct Node {
        int next[26] = {};
        int link = 0;
        vi word_ids;
    };
    vector<Node> t;
    vi bfs_order, pat_len;
    int num_patterns = 0;
    AhoCorasick() { t.emplace_back(); }
    int insert(const string& s) {
        int v = 0, id = num_patterns++;
        pat_len.pb(s.size());
        for (char ch : s) {

```

```

    int c = ch - 'a';
    if (!t[v].next[c]) t[v].next[c] = t.size(), t.
    emplace_back();
    v = t[v].next[c];
}
t[v].word_ids.pb(id);
return id;
}
void build() {
    queue<int> q;
    for (int c = 0; c < 26; c++) {
        if (t[0].next[c]) q.push(t[0].next[c]);
        else t[0].next[c] = 0;
    }
    while (!q.empty()) {
        int u = q.front(); q.pop();
        bfs_order.pb(u);
        for (int c = 0; c < 26; c++) {
            int v = t[u].next[c];
            if (v) t[v].link = t[t[u].link].next[c], q.push(
            v);
            else t[u].next[c] = t[t[u].link].next[c];
        }
    }
}
// Numero de ocurrencias de cada patron en text
vi count_occurrences(const string& text) {
    vi freq(t.size(), 0);
    int u = 0;
    for (char ch : text) u = t[u].next[ch - 'a'], freq[u]
    ++;
    for (int i = bfs_order.size() - 1; i >= 0; i--) {
        int cur = bfs_order[i];
        freq[t[cur].link] += freq[cur];
    }
    vi ans(num_patterns, 0);
    for (int i = 1; i < t.size(); i++) if (freq[i]) for
    (int id : t[i].word_ids) ans[id] += freq[i];
    return ans;
}
// Posicion de la primera ocurrencia de cada patron (o
-1)
vi first_occurrences(const string& text) {
    vi first(t.size(), INF);
    int u = 0;
    for (int i = 0; i < text.size(); i++) {
        u = t[u].next[text[i] - 'a'];
        first[u] = min(first[u], i);
    }
    for (int i = bfs_order.size() - 1; i >= 0; i--) {
        int cur = bfs_order[i];
        first[t[cur].link] = min(first[t[cur].link], first
        [cur]);
    }
    vi ans(num_patterns, -1);
    for (int i = 1; i < t.size(); i++) if (first[i] !=
    INF)
        for (int id : t[i].word_ids) {
            int st = first[i] - pat_len[id] + 1;
            ans[id] = ans[id] == -1 ? st : min(ans[id], st);
        }
    return ans;
}

```

```

};
/**
 * Uso[0]: SuffixAutomaton sam(s); sam.
    distinct_substrings(); sam.count_occurrences(p);
    sam.first_occurrence(p);
 * Automata de sufijos: substrings distintos,
    ocurrencias y primera aparicion de un patron.
 * Complejidad: construccion  $O(N \cdot 26)$ , consultas  $O(|p|)$ .
 */
struct SuffixAutomaton {
    struct Node {
        int len = 0, link = -1, next[26] = {};
        int firstpos = -1;
        ll cnt = 0;
    };
    vector<Node> t;
    int last;
    bool occ_computed = false;
    SuffixAutomaton(const string& s = "") {
        t.reserve(s.size() * 2 + 1);
        t.emplace_back();
        last = 0;
        for (char c : s) extend(c - 'a');
    }
    void extend(int c) {
        int cur = t.size();
        t.emplace_back();
        t[cur].len = t[last].len + 1;
        t[cur].firstpos = t[cur].len - 1;
        t[cur].cnt = 1;
        int p = last;
        while (p != -1 && !t[p].next[c]) t[p].next[c] = cur,
        p = t[p].link;
        if (p == -1) t[cur].link = 0;
        else {
            int q = t[p].next[c];
            if (t[p].len + 1 == t[q].len) t[cur].link = q;
            else {
                int clone = t.size();
                t.push_back(t[q]);
                t[clone].len = t[p].len + 1;
                t[clone].cnt = 0;
                while (p != -1 && t[p].next[c] == q) t[p].next[c]
                = clone, p = t[p].link;
                t[q].link = t[cur].link = clone;
            }
        }
        last = cur;
    }
    void compute_occurrences() {
        if (occ_computed) return;
        occ_computed = true;
        int n = t.size();
        vi c(n + 1, 0), order(n);
        for (int i = 0; i < n; i++) c[t[i].len]++;
        for (int i = 1; i <= n; i++) c[i] += c[i - 1];
        for (int i = 0; i < n; i++) order[--c[t[i].len]] = i;
        for (int i = n - 1; i > 0; i--) {
            int u = order[i];
            if (t[u].link != -1) t[t[u].link].cnt += t[u].cnt;
        }
    }
}

```

```

}
ll distinct_substrings() {
    ll ans = 0;
    for (int i = 1; i < t.size(); i++) ans += t[i].len -
    t[t[i].link].len;
    return ans;
}
int first_occurrence(const string& p) {
    int u = 0;
    for (char ch : p) {
        int c = ch - 'a';
        if (!t[u].next[c]) return -1;
        u = t[u].next[c];
    }
    return t[u].firstpos - p.size() + 1;
}
ll count_occurrences(const string& p) {
    compute_occurrences();
    int u = 0;
    for (char ch : p) {
        int c = ch - 'a';
        if (!t[u].next[c]) return 0;
        u = t[u].next[c];
    }
    return t[u].cnt;
}
}
/**
 * Uso[0]: SuffixArray sa(s); sa.p (sufijos ordenados);
    sa.get_lcp(i, j);
 * Arreglo de sufijos con tabla RMQ de lcp para
    consultar el prefijo comun mas largo de dos sufijos
    en  $O(1)$ .
 * Complejidad: construccion  $O(N \log N)$ , consulta lcp
     $O(1)$ .
 */
struct SuffixArray {
    string s;
    int n, log_n;
    vi p, c, lcp, rank;
    vector<vi> st;
    SuffixArray(string s) : s(s + "$"), n(s.size()) {
        build_sa(); build_lcp(); build_rmq();
    }
    void build_sa() {
        const int alphabet = 256;
        p.assign(n, 0); c.assign(n, 0);
        vi cnt(max(alphabet, n), 0), p_new(n), c_new(n);
        for (int i = 0; i < n; i++) cnt[s[i]]++;
        for (int i = 1; i < alphabet; i++) cnt[i] += cnt[i -
        1];
        for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;
        c[p[0]] = 0;
        int classes = 1;
        for (int i = 1; i < n; i++) {
            if (s[p[i]] != s[p[i - 1]]) classes++;
            c[p[i]] = classes - 1;
        }
        for (int k = 0; (1 << k) < n; k++) {
            for (int i = 0; i < n; i++) {
                p_new[i] = p[i] - (1 << k);
                if (p_new[i] < 0) p_new[i] += n;
            }
        }
    }
}

```

```

    }
    fill(cnt.begin(), cnt.begin() + classes, 0);
    for (int i = 0; i < n; i++) cnt[c[p_new[i]]]++;
    for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
    for (int i = n - 1; i >= 0; i--) p[--cnt[c[p_new[i]]]] = p_new[i];
    c_new[p[0]] = 0;
    classes = 1;
    for (int i = 1; i < n; i++) {
        pair<int, int> cur = {c[p[i]], c[(p[i] + (1 << k)) % n]};
        pair<int, int> prev = {c[p[i - 1]], c[(p[i - 1] + (1 << k)) % n]};
        if (cur != prev) classes++;
        c_new[p[i]] = classes - 1;
    }
    c.swap(c_new);
}

void build_lcp() {
    lcp.assign(n, 0);
    rank.assign(n, 0);
    for (int i = 0; i < n; i++) rank[p[i]] = i;
    int k = 0;
    for (int i = 0; i < n - 1; i++) {
        int pi = rank[i], j = p[pi - 1];
        while (s[i + k] == s[j + k]) k++;
        lcp[pi] = k;
        if (k) k--;
    }
}

void build_rmq() {
    log_n = log2(n) + 1;
    st.assign(n, vi(log_n));
    for (int i = 0; i < n; i++) st[i][0] = lcp[i];
    for (int j = 1; j < log_n; j++)
        for (int i = 0; i + (1 << j) <= n; i++)
            st[i][j] = min(st[i][j - 1], st[i + (1 << (j - 1))][j - 1]);
}

// LCP de Los sufijos que empiezan en i y j
int get_lcp(int i, int j) {
    if (i == j) return n - 1 - i;
    int u = rank[i], v = rank[j];
    if (u > v) swap(u, v);
    u++;
    int k = log2(v - u + 1);
    return min(st[u][k], st[v - (1 << k) + 1][k]);
}

/**
 * Uso[0]: PalindromicTree pt(s); pt.count_palindromes();
 * pt.compute_occurrences();
 * Arbol de palindromos (eertree): todos Los palindromos
 * distintos, sus ocurrencias y su primera posicion.
 * Complejidad: construccion  $O(N \cdot 26)$ , ocurrencias  $O(N)$ 
 */
struct PalindromicTree {
    struct Node {
        int next[26] = {};

```

```

        int len = 0, link = 0;
        ll occ = 0;
        int first_pos = -1; // posicion donde aparece por
        // primera vez
    };
    vector<Node> t;
    string s; // s[0] = sentinel, nunca coincide con una
    // letra
    int last;
    PalindromicTree(const string& str = "") {
        t.emplace_back(); t.emplace_back();
        t[0].len = -1; t[0].link = 0; // raiz imaginaria (
        // len -1)
        t[1].len = 0; t[1].link = 0; // raiz del vacio (len
        // 0)
        last = 1;
        s = "$";
        for (char c : str) add(c);
    }

    int get_link(int v, int pos) {
        while (true) {
            int idx = pos - t[v].len - 1;
            if (idx >= 0 && s[idx] == s[pos]) break;
            v = t[v].link;
        }
        return v;
    }

    void add(char c) {
        s += c;
        int pos = s.size() - 1;
        int cur = get_link(last, pos);
        int idx = c - 'a';
        if (!t[cur].next[idx]) {
            int now = t.size();
            t.emplace_back();
            t[now].len = t[cur].len + 2;
            t[now].first_pos = pos - t[now].len; // 0-indexado
            t[now].link = t[now].len == 1 ? 1 : t[get_link(t[
            cur].link, pos)].next[idx];
            t[cur].next[idx] = now;
        }
        last = t[cur].next[idx];
        t[last].occ++;
    }

    ll count_palindromes() { return t.size() - 2; } //
    // palindromos distintos
    void compute_occurrences() { // occ[u] = veces que
    // aparece el palindromo u
        for (int i = t.size() - 1; i > 1; i--) t[t[i].link].
        occ += t[i].occ;
    }
};

```

Dynamic Programming

```

/**
 * Uso: LiChaoTree cht; cht.add(m, c); ll v = cht.query(
 * x);
 * Arbol de Li Chao para consultar el minimo de  $m \cdot x + c$ 
 * entre Las rectas insertadas (cambiar < por > para
 * max).
 * Complejidad: insercion y consulta  $O(\log R)$  con R el
 * tamaño del rango de x.
 */

```

```

*/
struct LiChaoTree {
    struct Line {
        ll m, c;
        Line(ll m = 0, ll c = LINF) : m(m), c(c) {}
        ll eval(ll x) const { return m * x + c; }
    };
    struct Node { Line line; int lc = -1, rc = -1; };
    vector<Node> st;
    ll min_x, max_x;
    LiChaoTree(ll mn = -1e9, ll mx = 1e9) : min_x(mn),
    max_x(mx) { st.emplace_back(); }
    void add_line(Line nw, int v, ll l, ll r) {
        ll mid = l + (r - l) / 2;
        bool left = nw.eval(l) < st[v].line.eval(l);
        bool midb = nw.eval(mid) < st[v].line.eval(mid);
        if (midb) swap(st[v].line, nw);
        if (l == r) return;
        if (left != midb) {
            if (st[v].lc == -1) st[v].lc = st.size(), st.
            emplace_back();
            add_line(nw, st[v].lc, l, mid);
        } else {
            if (st[v].rc == -1) st[v].rc = st.size(), st.
            emplace_back();
            add_line(nw, st[v].rc, mid + 1, r);
        }
    }

    void add(ll m, ll c) { add_line(Line(m, c), 0, min_x,
    max_x); }

    ll query(ll x, int v, ll l, ll r) {
        if (v == -1) return LINF;
        ll res = st[v].line.eval(x);
        if (l == r) return res;
        ll mid = l + (r - l) / 2;
        if (x <= mid) return min(res, query(x, st[v].lc, l,
        mid));
        return min(res, query(x, st[v].rc, mid + 1, r));
    }

    ll query(ll x) { return query(x, 0, min_x, max_x); }
};

/**
 * Uso: dnc_dp(1, n, 0, n, dp_prev, dp_curr, cost);
 * Divide & conquer DP: resuelve
 *  $dp[i][j] = \min_{k < j} (dp[i - 1][k] + cost(k + 1, j))$  cuando La
 * decision optima es monotona.
 * Complejidad:  $O(KN \log N)$  con cost evaluada en  $O(1)$ .
 */
template<class CostFunc>
void dnc_dp(int l, int r, int optl, int optr, const vll&
    prev, vll& curr, CostFunc& cost) {
    if (l > r) return;
    int mid = (l + r) / 2;
    pair<ll, int> best = {LINF, -1};
    for (int k = optl; k <= min(mid - 1, optr); k++) {
        ll cur = prev[k] == LINF ? LINF : prev[k] + cost(k +
        1, mid);
        if (cur < best.fi) best = {cur, k};
    }
    curr[mid] = best.fi;
    int opt = best.se == -1 ? optl : best.se;
    dnc_dp(l, mid - 1, optl, opt, prev, curr, cost);
}

```

```

    dnc_dp(mid + 1, r, opt, optr, prev, curr, cost);
}

/**
 * Uso: auto dp = knuth_dp(n, cost);
 * Optimizacion de Knuth para DP de intervalos
 *  $dp[i][j] = \min_k (dp[i][k] + dp[k][j]) + cost(i, j)$  con decision monotona.
 * Complejidad:  $O(N^2)$ .
 */
template<class CostFunc>
vector<vll> knuth_dp(int n, CostFunc& cost) {
    vector<vll> dp(n + 1, vll(n + 1, 0));
    vector<vi> opt(n + 1, vi(n + 1, 0));
    for (int i = 0; i < n; i++) opt[i][i + 1] = i;
    for (int len = 2; len <= n; len++)
        for (int i = 0; i + len <= n; i++) {
            int j = i + len;
            dp[i][j] = LINF;
            for (int k = opt[i][j - 1]; k <= min(j - 1, opt[i + 1][j]); k++) {
                ll cur = dp[i][k] + dp[k][j] + cost(i, j);
                if (cur < dp[i][j]) dp[i][j] = cur, opt[i][j] = k;
            }
        }
    return dp;
}

/**
 * Uso: DigitDP::solve(R) - DigitDP::solve(L-1);
 * DP sobre digitos para contar numeros en  $[0, N]$  que cumplan condiciones; ajustar memo y transiciones segun el problema.
 * Complejidad:  $O(\text{digitos} \times \text{estados})$  por llamada.
 */
namespace DigitDP {
    string s;
    ll memo[20][2][2]; // agregar estados segun problema
    ll dp(int idx, bool tight, bool lz) {
        if (idx == s.size()) return 1; // 1 si el numero es valido
        if (memo[idx][tight][lz] != -1) return memo[idx][tight][lz];
        int lim = tight ? s[idx] - '0' : 9;
        ll ans = 0;
        for (int d = 0; d <= lim; d++)
            ans += dp(idx + 1, tight && d == lim, lz && d == 0);
        return memo[idx][tight][lz] = ans;
    }
    ll solve(ll n) {
        if (n < 0) return 0;
        s = to_string(n);
        memset(memo, -1, sizeof(memo));
        return dp(0, true, true);
    }
}

/**
 * Uso: sos_dp(dp, n_bits);
 * Suma sobre subconjuntos:  $dp[mask]$  acumula la suma de todos sus subconjuntos (sum over subsets).

```

```

 * Complejidad:  $O(N \cdot 2^N)$ .
 */
void sos_dp(vll& dp, int n) {
    for (int i = 0; i < n; i++)
        for (int mask = 0; mask < (1 << n); mask++)
            if (mask & (1 << i)) dp[mask] += dp[mask ^ (1 << i)];
}

/**
 * Uso: auto [len, seq] = lis(a);
 * Subsecuencia creciente mas larga estricta, con reconstruccion de la secuencia.
 * Complejidad:  $O(N \log N)$ .
 */
pair<int, vi> lis(const vi& a) {
    int n = a.size();
    vi d, pos, parent(n, -1);
    for (int i = 0; i < n; i++) {
        auto it = lower_bound(all(d), a[i]);
        int idx = it - d.begin();
        if (it == d.end()) d.pb(a[i]), pos.pb(i);
        else *it = a[i], pos[idx] = i;
        if (idx > 0) parent[i] = pos[idx - 1];
    }
    vi seq;
    for (int cur = pos.empty() ? -1 : pos.back(); cur != -1; cur = parent[cur]) seq.pb(a[cur]);
    reverse(all(seq));
    return {d.size(), seq};
}

/**
 * Uso: bitset_knapsack(pesos, target);
 * Determina si un subconjunto de pesos suma exactamente un valor objetivo mediante bitsets.
 * Complejidad:  $O(N \cdot S/64)$  con  $S$  la suma maxima ( $10^5$  fija en MAX_SUM).
 */
const int MAX_SUM = 1e5 + 5;
bool bitset_knapsack(const vi& weights, int target) {
    bitset<MAX_SUM> dp;
    dp[0] = 1;
    for (int w : weights) dp |= dp << w;
    return dp[target];
}

/**
 * Uso: LL ans = WQS::solve(check, K);
 * Busqueda binaria de Lagrange (Alien's trick) para optimizar un DP con la restricción de usar exactamente  $K$  elementos o particiones.
 * Complejidad:  $O(\log R \cdot f)$  con  $f$  el costo del DP por evaluacion y  $R$  el rango de la penalidad.
 */
namespace WQS {
    template <class F>
    ll solve(F check, int K) {
        ll lo = -LINF, hi = LINF;
        while (lo + 1 < hi) {
            ll mid = lo + (hi - lo) / 2;
            if (check(mid).se >= K) lo = mid;
            else hi = mid;
        }
    }
}

```

```

}
auto [c, k] = check(lo);
return c - lo * (ll)K; // quita la penalidad de los K elementos
}

/**
 * Uso: CHTDeque cht; cht.add(m, c); cht.query(x); // y cht.query_bs(x) para x arbitraria
 * Convex hull trick con deque para minimos  $m \cdot x + c$  con pendientes monotonas y consultas con  $x$  creciente.
 * Complejidad: insercion y consulta amortizadas  $O(1)$ ; consulta binaria  $O(\log N)$ .
 */
struct CHTDeque {
    struct Line { ll m, c; };
    deque<Line> dq;
    int slope_inc = 0; // +1 pendientes crecientes, -1 decrecientes (auto-detectado)
    static ll eval(const Line& l, ll x) { return l.m * x + l.c; }
    // true si l2 queda obsoleta entre l1 y l3
    bool is_bad(const Line& l1, const Line& l2, const Line& l3) {
        if (slope_inc > 0) return (__int128)(l1.c - l2.c) * (l3.m - l2.m) <= (__int128)(l2.c - l3.c) * (l2.m - l1.m);
        return (__int128)(l3.c - l1.c) * (l1.m - l2.m) <= (__int128)(l2.c - l1.c) * (l1.m - l3.m);
    }
    void add(ll m, ll c) {
        if (!dq.empty() && dq.back().m == m) {
            if (dq.back().c <= c) return; // la nueva no aporta (min)
            dq.pop_back();
        }
        if (dq.size() >= 2 && slope_inc == 0) slope_inc = dq.back().m < m ? 1 : -1;
        Line nw{m, c};
        while (dq.size() >= 2 && is_bad(dq[dq.size() - 2], dq.back(), nw)) dq.pop_back();
        dq.push_back(nw);
    }
    ll query(ll x) { // x debe ser no decreciente
        if (slope_inc > 0) { // pendientes crecientes: optimo va del fondo al frente
            while (dq.size() >= 2 && eval(dq.back(), x) >= eval(dq[dq.size() - 2], x)) dq.pop_back();
            return eval(dq.back(), x);
        }
        while (dq.size() >= 2 && eval(dq.front(), x) >= eval(dq[1], x)) dq.pop_front();
        return eval(dq.front(), x);
    }
    ll query_bs(ll x) { // x arbitraria
        int lo = 0, hi = (int)dq.size() - 1;
        while (lo < hi) {
            int mid = (lo + hi) / 2;
            if (eval(dq[mid], x) <= eval(dq[mid + 1], x)) hi = mid;
            else lo = mid + 1;
        }
    }
}

```

```

    return eval(dq[lo], x);
}
};

```

Mathematics

```

namespace MathAlgo {
/**
 * Uso: modpow(a, b, mod), modinv(a, mod), ext_gcd(a,
 * b, x, y);
 * Potencia modular, inverso modular y euclides
 * extendido ( $ax + by = \gcd(a, b)$ ).
 * Complejidad:  $O(\log b)$  por operacion.
 */
ll modpow(ll a, ll b, ll m = MOD) {
    ll r = 1; a %= m;
    while (b) { if (b & 1) r = r * a % m; a = a * a % m;
        b >>= 1; }
    return r;
}
// Euclides extendido: resuelve  $ax + by = \gcd(a, b)$ 
ll ext_gcd(ll a, ll b, ll& x, ll& y) {
    if (!b) { x = 1; y = 0; return a; }
    ll x1, y1, d = ext_gcd(b, a % b, x1, y1);
    x = y1; y = x1 - y1 * (a / b); return d;
}
/**
 * Uso: auto [x, lcm] = modinv(a, mod);
 * Inverso modular de a modulo m (exige  $\gcd(a, m) = 1$ ).
 * Complejidad:  $O(\log m)$ .
 */
ll modinv(ll a, ll m = MOD) {
    ll x, y; assert(ext_gcd(a, m, x, y) == 1);
    return (x % m + m) % m;
}
/**
 * Uso: auto [x, lcm] = crt(a1, m1, a2, m2);
 * Teorema chino del resto para dos congruencias;
 * devuelve {-1, -1} si no tienen solucion.
 * Complejidad:  $O(\log \min(m_1, m_2))$ .
 */
pair<ll, ll> crt(ll a1, ll m1, ll a2, ll m2) {
    ll p, q, g = ext_gcd(m1, m2, p, q);
    if (a1 % g != a2 % g) return {-1, -1};
    ll lcm = (m1 / g) * m2;
    ll res = a1 + (ll)((__int128_t)(a2 - a1) / g * p % (
        m2 / g)) * m1;
    return {(res % lcm + lcm) % lcm, lcm};
}
/**
 * Uso: auto [x, lcm] = solve_crt_array(A, M);
 * Teorema chino del resto para un arreglo de
 * congruencias; devuelve {-1, -1} si alguna es
 * inconsistente.
 * Complejidad:  $O(K \log m)$  con K el numero de
 * congruencias.
 */
pair<ll, ll> solve_crt_array(const vector<ll>& A,
    const vector<ll>& M) {
    ll ans = A[0];
    ll lcm = M[0];
    for (int i = 1; i < A.size(); i++) {

```

```

        auto res = crt(ans, lcm, A[i], M[i]);
        if (res.first == -1) return {-1, -1}; // No
        hay solución
        ans = res.first;
        lcm = res.second;
    }
    return {ans, lcm};
}
/**
 * Uso: Sieve sv(n); sv.primes; sv.spf; sv.mu; sv.phi;
 * sv.factorize(x);
 * Criba lineal con lista de primos, menor factor
 * primo, funcion de Mobius y totient de Euler.
 * Complejidad: criba  $O(N)$ , factorizacion de un valor
 *  $O(\log x)$ .
 */
struct Sieve {
    vector<int> primes, spf, mu, phi;
    Sieve(int n) {
        spf.assign(n + 1, 0); mu.assign(n + 1, 0); phi.
        assign(n + 1, 0);
        mu[1] = phi[1] = 1;
        for (int i = 2; i <= n; i++) {
            if (!spf[i]) spf[i] = i, primes.pb(i), mu[i] =
                -1, phi[i] = i - 1;
            for (int p : primes) {
                if (p > spf[i] || 1LL * i * p > n) break;
                spf[i * p] = p;
                if (i % p == 0) { mu[i * p] = 0; phi[i * p] =
                    phi[i] * p; break; }
                mu[i * p] = -mu[i]; phi[i * p] = phi[i] * (p -
                    1);
            }
        }
    }
    vector<int> factorize(int x) { // factores primos
        con multiplicidad
        vector<int> r;
        while (x > 1) r.pb(spf[x]), x /= spf[x];
        return r;
    }
};
/**
 * Uso: Combinatorics c(n, mod); c.nCr(n, r); c.
 * catalan(n);
 * Factoriales e inversos para combinar y numeros de
 * Catalan modulo primo.
 * Complejidad: preproceso  $O(N)$ , consulta  $O(1)$ .
 */
struct Combinatorics {
    vector<ll> fact, invfact;
    Combinatorics(int n, ll mod = MOD) {
        fact.assign(n + 1, 1); invfact.assign(n + 1, 1);
        for (int i = 1; i <= n; i++) fact[i] = fact[i - 1]
            * i % mod;
        invfact[n] = modinv(fact[n], mod);
        for (int i = n - 1; i >= 0; i--) invfact[i] =
            invfact[i + 1] * (i + 1) % mod;
        this->mod = mod;
    }
    ll mod;

```

```

    ll nCr(int n, int r) {
        if (r < 0 || r > n) return 0;
        return fact[n] * invfact[r] % mod * invfact[n - r]
            % mod;
    }
    ll catalan(int n) { return nCr(2 * n, n) * modinv(n
        + 1, mod) % mod; }
};
/**
 * Uso: gauss(matriz_extendida, soluciones);
 * Eliminacion gaussiana con pivoteo para sistemas
 * lineales; devuelve 1, INF o 0 segun el numero de
 * soluciones.
 * Complejidad:  $O(n \cdot m^2)$ .
 */
int gauss(vector<vector<double>>& a, vector<double>&
    ans) {
    int n = a.size(), m = a[0].size() - 1;
    vector<int> where(m, -1);
    for (int col = 0, row = 0; col < m && row < n; col
        ++){
        int sel = row;
        for (int i = row; i < n; i++) if (fabs(a[i][col])
            > fabs(a[sel][col])) sel = i;
        if (fabs(a[sel][col]) < EPS) continue;
        for (int i = col; i <= m; i++) swap(a[sel][i], a[
            row][i]);
        where[col] = row;
        for (int i = 0; i < n; i++) if (i != row) {
            double c = a[i][col] / a[sel][col];
            for (int j = col; j <= m; j++) a[i][j] -= a[sel
                ][j] * c;
        }
        row++;
    }
    ans.assign(m, 0);
    for (int i = 0; i < m; i++) if (where[i] != -1) ans[
        i] = a[where[i]][m] / a[where[i]][i];
    for (int i = 0; i < n; i++) {
        double s = 0;
        for (int j = 0; j < m; j++) s += ans[j] * a[i][j];
        if (fabs(s - a[i][m]) > EPS) return 0;
    }
    for (int i = 0; i < m; i++) if (where[i] == -1)
        return INF;
    return 1;
}
/**
 * Uso: fft(vec_complejos, invert); auto c = multiply(
 * a, b);
 * Transformada rapida de Fourier iterativa y
 * multiplicacion exacta de polinomios con
 * coeficientes enteros.
 * Complejidad:  $O(N \log N)$  con N potencia de dos.
 */
using cd = complex<double>;
// FFT iterativa, invert = true para transformada
// inversa
void fft(vector<cd>& a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {

```

```

    int bit = n >> 1;
    for (; j & bit; bit >>= 1) j ^= bit;
    j ^= bit;
    if (i < j) swap(a[i], a[j]);
}
for (int len = 2; len <= n; len <= 1) {
    double ang = 2 * acos(-1.0) / len * (invert ? -1 : 1);
    cd wlen(cos(ang), sin(ang));
    for (int i = 0; i < n; i += len) {
        cd w(1);
        for (int j = 0; j < len / 2; j++) {
            cd u = a[i + j], v = a[i + j + len / 2] * w;
            a[i + j] = u + v; a[i + j + len / 2] = u - v;
            w *= wlen;
        }
    }
    if (invert) for (cd& x : a) x /= n;
}
// Multiplicacion de polinomios (coeficientes enteros)
// O(n log n)
vector<int> multiply(vector<int> const& a, vector<int>
    const& b) {
    vector<cd> fa(all(a)), fb(all(b));
    int n = 1;
    while (n < a.size() + b.size()) n <= 1;
    fa.resize(n); fb.resize(n);
    fft(fa, false); fft(fb, false);
    for (int i = 0; i < n; i++) fa[i] *= fb[i];
    fft(fa, true);
    vector<int> res(n);
    for (int i = 0; i < n; i++) res[i] = round(fa[i].
        real());
    while (res.size() > 1 && !res.back()) res.pop_back()
        ;
    return res;
}
/**
 * Uso: ntt(vec, invert); auto c = multiply_mod(a, b);
 * Transformada numerica rapida (NTT) y multiplicacion
 * de polinomios con coeficientes modulares.
 * Complejidad: O(N log N) con N potencia de dos.
 */
void ntt(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <= 1) {
        ll wlen = modpow(3, (MOD - 1) / len);
        if (invert) wlen = modinv(wlen);
        for (int i = 0; i < n; i += len) {
            ll w = 1;
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2] * w %
                    MOD;
                a[i + j] = (u + v) % MOD;
                a[i + j + len / 2] = (u - v + MOD) % MOD;
            }
        }
    }
}

```

```

        w = w * wlen % MOD;
    }
}
if (invert) {
    ll ninv = modinv(n);
    for (ll& x : a) x = x * ninv % MOD;
}
}
vector<ll> multiply_mod(vector<ll> const& a, vector<ll>
    > const& b) {
    vector<ll> fa(all(a)), fb(all(b));
    int n = 1;
    while (n < a.size() + b.size()) n <= 1;
    fa.resize(n); fb.resize(n);
    ntt(fa, false); ntt(fb, false);
    for (int i = 0; i < n; i++) fa[i] = fa[i] * fb[i] %
        MOD;
    ntt(fa, true);
    while (fa.size() > 1 && !fa.back()) fa.pop_back();
    return fa;
}
/**
 * Uso: nim_game(pilas), sum_n(n), sum_squares(n),
 * geom_sum(a, r, n);
 * Decide el ganador del juego de Nim y calcula
 * sumatorias cerradas (aritmetica, cuadrados y
 * geometrica) modulo M.
 * Complejidad: O(N) para Nim, O(log M) para las
 * sumatorias.
 */
bool nim_game(const vi& piles) {
    int x = 0;
    for (int p : piles) x ^= p;
    return x != 0;
}
ll sum_n(ll n, ll m = MOD) { n %= m; return n * (n +
    1) % m * modinv(2, m) % m; }
ll sum_squares(ll n, ll m = MOD) {
    n %= m;
    return n * (n + 1) % m * (2 * n + 1) % m * modinv(6,
        m) % m;
}
ll geom_sum(ll a, ll r, ll n, ll m = MOD) {
    if (r == 1) return (n % m) * (a % m) % m;
    ll num = (modpow(r, n, m) - 1 + m) % m;
    return (a % m) * num % m * modinv((r - 1 + m) % m, m)
        % m;
}
/**
 * Uso: Matrix m(r, c); m.mat; m * o; m.power(p);
 * Matrix::identity(n);
 * Matriz con multiplicacion modular y exponenciacion
 * para recurrencias lineales y conteo de caminos.
 * Complejidad: producto O(r · c²), potencia O(r³ log p).
 */
struct Matrix {
    int r, c;
    vector<vector<ll>> mat;
    Matrix(int r, int c) : r(r), c(c), mat(r, vector<ll>
        >(c, 0)) {}
}

```

```

static Matrix identity(int n) {
    Matrix I(n, n);
    for (int i = 0; i < n; i++) I.mat[i][i] = 1;
    return I;
}
Matrix operator*(const Matrix& o) const {
    Matrix res(r, o.c);
    for (int i = 0; i < r; i++)
        for (int k = 0; k < c; k++) if (mat[i][k])
            for (int j = 0; j < o.c; j++)
                res.mat[i][j] = (res.mat[i][j] + mat[i][k] *
                    o.mat[k][j]) % MOD;
    return res;
}
Matrix power(ll p) const {
    Matrix res = identity(r), base = *this;
    while (p) { if (p & 1) res = res * base; base =
        base * base; p >>= 1; }
    return res;
}
};
/**
 * Uso: auto C = berlekamp_massey(sequencia);
 * Encuentra la recurrencia lineal minima de una
 * secuencia; C[0] · S[n] + C[1] · S[n - 1] + ... = 0.
 * Complejidad: O(N²).
 */
vector<ll> berlekamp_massey(const vector<ll>& s) {
    vector<ll> C{1}, B{1};
    int L = 0; ll m = 1, b = 1;
    for (int i = 0; i < s.size(); i++) {
        ll d = 0;
        for (int j = 0; j <= L; j++) d = (d + C[j] * s[i -
            j]) % MOD;
        if (d) {
            vector<ll> T = C;
            ll coef = MOD - d * modpow(b, MOD - 2) % MOD;
            while (C.size() <= B.size() + m) C.pb(0);
            for (int j = 0; j < B.size(); j++) C[j + m] = (C
                [j + m] + coef * B[j]) % MOD;
            if (2 * L <= i) L = i + 1 - L, B = T, b = d, m =
                1;
            else m++;
        } else m++;
    }
    C.resize(L + 1);
    return C;
}
/**
 * Uso: PollardRho::is_prime(n); PollardRho::factorize
 * (n);
 * Test de primalidad Miller-Rabin y factorizacion de
 * enteros de hasta 2⁶⁴ con Pollard-Rho.
 * Complejidad: O(n¹/⁴) esperado por factor encontrado
 * .
 */
struct PollardRho {
    using u64 = uint64_t; using u128 = __uint128_t;
    static u64 modpow(u64 b, u64 e, u64 m) {
        u64 r = 1; b %= m;
        while (e) { if (e & 1) r = (u128)r * b % m; b = (

```

```

u128)b * b % m; e >= 1; }
return r;
}
static bool is_prime(u64 n) {
    if (n < 2) return false;
    for (u64 p : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
31, 37}) {
        if (n == p) return true;
        if (n % p == 0) return false;
    }
    u64 d = n - 1; int s = 0;
    while (!(d & 1)) d >>= 1, s++;
    for (u64 a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29,
31, 37}) {
        if (a >= n) continue;
        u64 x = modpow(a, d, n);
        if (x == 1 || x == n - 1) continue;
        bool comp = true;
        for (int r = 1; r < s; r++) {
            x = (u128)x * x % n;
            if (x == n - 1) { comp = false; break; }
        }
        if (comp) return false;
    }
    return true;
}
static u64 get_factor(u64 n) {
    if (!(n & 1)) return 2;
    if (is_prime(n)) return n;
    u64 x = 2, y = 2, d = 1, c = 1;
    auto f = [&](u64 x) { return (u64)((u128)x * x %
n + c) % n; };
    while (d == 1) {
        x = f(x); y = f(f(y));
        d = gcd(x > y ? x - y : y - x, n);
        if (d == n) { x = rand() % (n - 2) + 2; y = x; c
= rand() % (n - 1) + 1; d = 1; }
    }
    return d;
}
static void factor(u64 n, map<u64, int>& f) {
    if (n == 1) return;
    if (is_prime(n)) { f[n]++; return; }
    u64 d = get_factor(n);
    factor(d, f); factor(n / d, f);
}
static map<u64, int> factorize(u64 n) { // {primo,
exponente}
    map<u64, int> f;
    factor(n, f);
    return f;
}
}
/**
 * Uso: Lucas l(p); l.solve(n, r);
 * Combinaciones modulo un primo pequeno con n y r
potencialmente enormes (teorema de Lucas).
 * Complejidad: O(p) de preproceso, O(logp n) por
consulta.
 */
struct Lucas {
    Combinatorics comb;

```

```

ll P;
Lucas(ll p) : P(p), comb(p - 1, p) {}
ll solve(ll n, ll r) {
    if (r < 0 || r > n) return 0;
    if (!r) return 1;
    if (n % P < r % P) return 0;
    return solve(n / P, r / P) * comb.nCr(n % P, r % P
) % P;
}
}
/**
 * Uso: fwht_xor(v, invert); auto c = xor_convolution(
a, b); // igual para or_ y and_
 * Transformada de Walsh-Hadamard y convoluciones
sobre operaciones de bits (xor, and, or) modulo M.
 * Complejidad: O(N log N).
 */
void fwht_xor(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int len = 2; len <= n; len <= 1) {
        for (int i = 0; i < n; i += len) {
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2];
                a[i + j] = (u + v) % MOD;
                a[i + j + len / 2] = (u - v + MOD) % MOD;
            }
        }
    }
    if (invert) {
        ll inv = modinv(n);
        for (ll& x : a) x = x * inv % MOD;
    }
}
void fwht_or(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int len = 2; len <= n; len <= 1) {
        for (int i = 0; i < n; i += len) {
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2];
                if (invert) a[i + j + len / 2] = (v - u + MOD)
% MOD;
                else a[i + j + len / 2] = (u + v) % MOD;
            }
        }
    }
}
void fwht_and(vector<ll>& a, bool invert) {
    int n = a.size();
    for (int len = 2; len <= n; len <= 1) {
        for (int i = 0; i < n; i += len) {
            for (int j = 0; j < len / 2; j++) {
                ll u = a[i + j], v = a[i + j + len / 2];
                if (invert) a[i + j] = (u - v + MOD) % MOD;
                else a[i + j] = (u + v) % MOD;
            }
        }
    }
}
vector<ll> xor_convolution(vector<ll> a, vector<ll> b)
{
    int n = 1;
    while (n < max(a.size(), b.size())) n <= 1;

```

```

a.resize(n); b.resize(n);
fwht_xor(a, false); fwht_xor(b, false);
for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % MOD
;
fwht_xor(a, true);
return a;
}
vector<ll> or_convolution(vector<ll> a, vector<ll> b)
{
    int n = 1;
    while (n < max(a.size(), b.size())) n <= 1;
    a.resize(n); b.resize(n);
    fwht_or(a, false); fwht_or(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % MOD
;
    fwht_or(a, true);
    return a;
}
vector<ll> and_convolution(vector<ll> a, vector<ll> b)
{
    int n = 1;
    while (n < max(a.size(), b.size())) n <= 1;
    a.resize(n); b.resize(n);
    fwht_and(a, false); fwht_and(b, false);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i] % MOD
;
    fwht_and(a, true);
    return a;
}
}
/**
 * Uso: floor_sum(n, m, a, b);
 * Calcula  $\sum_{i=0}^{n-1} \lfloor (a \cdot i + b) / m \rfloor$ .
 * Complejidad: O(log m).
 */
ll floor_sum(ll n, ll m, ll a, ll b) {
    ll ans = 0;
    while (true) {
        if (a >= m) { ans += (n - 1) * n * (a / m) / 2; a
%= m; }
        if (b >= m) { ans += n * (b / m); b %= m; }
        ll y_max = a * n + b;
        if (y_max < m) break;
        n = y_max / m;
        b = y_max % m;
        swap(m, a);
    }
    return ans;
}
}


## Geometry


/**
 * Uso: cmp(a, b);
 * Comparacion de numeros de punto flotante con
tolerancia EPS; devuelve -1, 0 o 1.
 * Complejidad: O(1).
 */
int cmp(ld x, ld y = 0) { return x <= y + EPS ? (x + EPS
< y ? -1 : 0) : 1; }
/**

```

```

* Uso: Point<ll> p(x, y); p+q, p-q, p*c, p/c, p==q;
orden lexicografico con <
* Punto 2D generico con operadores aritmeticos,
igualdad y orden lexicografico.
* Complejidad: O(1) por operacion.
*/
template<typename T> struct Point {
    T x, y;
    Point(T x = 0, T y = 0) : x(x), y(y) {}
    Point operator+(const Point& p) const { return {x + p.x, y + p.y}; }
    Point operator-(const Point& p) const { return {x - p.x, y - p.y}; }
    Point operator*(T c) const { return {x * c, y * c}; }
    Point operator/(T c) const { return {x / c, y / c}; }
    bool operator==(const Point& p) const { return x == p.x && y == p.y; }
    bool operator<(const Point& p) const { return x < p.x || (x == p.x && y < p.y); } // Lexicografico
};
/**
* Uso: dot(a, b), cross(a, b), cross(p, a, b), norm2(p), norm(p), sgn(v);
* Operaciones vectoriales basicas: producto punto, producto cruz, norma al cuadrado, norma y signo.
* Complejidad: O(1) por operacion.
*/
template<typename T> T dot(Point<T> a, Point<T> b) {
    return a.x * b.x + a.y * b.y; }
template<typename T> T cross(Point<T> a, Point<T> b) {
    return a.x * b.y - a.y * b.x; }
template<typename T> T cross(Point<T> p, Point<T> a, Point<T> b) { return cross(a - p, b - p); }
template<typename T> T norm2(Point<T> p) { return dot(p, p); }
template<typename T> T ld norm(Point<T> p) { return sqrtl(norm2(p)); }
template<typename T> int sgn(T(v)) { return (T(0) < v) - (v < T(0)); }
/**
* Uso: orient(a, b, c), on_segment(p, a, b), segment_intersect(a, b, c, d);
* Orientacion de tres puntos (horaria, colineal o antihoraria), pertenencia de un punto a un segmento e interseccion de segmentos.
* Complejidad: O(1) por operacion.
*/
template<typename T> int orient(Point<T> a, Point<T> b, Point<T> c) { return sgn(cross(a, b, c)); }

template<typename T> bool on_segment(Point<T> p, Point<T> a, Point<T> b) {
    return orient(a, b, p) == 0 && min(a.x, b.x) <= p.x && p.x <= max(a.x, b.x) && min(a.y, b.y) <= p.y && p.y <= max(a.y, b.y);
}

template<typename T> bool segment_intersect(Point<T> a, Point<T> b, Point<T> c, Point<T> d) {
    int o1 = orient(a, b, c), o2 = orient(a, b, d), o3 = orient(c, d, a), o4 = orient(c, d, b);
    if (o1 != o2 && o3 != o4) return true;
    if (o1 == 0 && on_segment(c, a, b)) return true;
    if (o2 == 0 && on_segment(d, a, b)) return true;
}

if (o3 == 0 && on_segment(a, c, d)) return true;
if (o4 == 0 && on_segment(b, c, d)) return true;
return false;
}
/**
* Uso: polygon_area_2(poly) / 2 = area, boundary_points(poly), point_in_polygon(poly, p);
* Area doble de un poligono (para evitar floats), puntos del reticulo en el borde (teorema de Pick) y clasificacion de un punto (0 fuera, 1 borde, 2 dentro).
* Complejidad: O(N) por operacion con N vertices.
*/
template<typename T> T polygon_area_2(const vector<Point<T>>& p) {
    T area = 0;
    for (int i = 0; i < p.size(); i++) area += cross(p[i], p[(i + 1) % p.size()]);
    return abs(area);
}
// Puntos del reticulo en el borde (Pick: 2*A = 2*I + B - 2)
long long boundary_points(const vector<Point<ll>>& poly) {
    ll b = 0;
    for (int i = 0; i < poly.size(); i++) {
        auto p1 = poly[i], p2 = poly[(i + 1) % poly.size()];
        b += gcd(abs(p1.x - p2.x), abs(p1.y - p2.y));
    }
    return b;
}
// Retorna 0 afuera, 1 en el borde, 2 adentro
template<typename T> int point_in_polygon(const vector<Point<T>>& poly, Point<T> p) {
    int hits = 0;
    for (int i = 0; i < poly.size(); i++) {
        auto a = poly[i], b = poly[(i + 1) % poly.size()];
        if (on_segment(p, a, b)) return 1;
        if (a.y > b.y) swap(a, b);
        if (p.y >= a.y && p.y < b.y && cross(p, a, b) > 0) hits++;
    }
    return hits & 1 ? 2 : 0;
}
/**
* Uso: auto hull = convex_hull(puntos);
* Cascara convexa (monotonic chain) en sentido antihorario, sin puntos colineales en el borde.
* Complejidad: O(N log N).
*/
template<typename T> vector<Point<T>> convex_hull(vector<Point<T>> pts) {
    int n = pts.size(), k = 0;
    if (n <= 2) return pts;
    vector<Point<T>> h(2 * n);
    sort(all(pts));
    for (int i = 0; i < n; i++) {
        while (k >= 2 && cross(h[k - 2], h[k - 1], pts[i]) <= 0) k--;
        h[k++] = pts[i];
    }
}

for (int i = n - 2, t = k + 1; i >= 0; i--) {
    while (k >= t && cross(h[k - 2], h[k - 1], pts[i]) <= 0) k--;
    h[k++] = pts[i];
}
h.resize(k - 1);
return h;
}
/**
* Uso: closest_pair(puntos);
* Distancia al cuadrado entre el par de puntos mas cercanos.
* Complejidad: O(N log N).
*/
long long closest_pair(vector<Point<ll>> pts) {
    int n = pts.size();
    if (n < 2) return 8e18;
    sort(all(pts));
    set<pair<ll, ll>> active; // {y, x}
    ll best = 8e18;
    int left = 0;
    for (int i = 0; i < n; i++) {
        ll d = ceil(sqrtl(best));
        while (left < i && pts[i].x - pts[left].x >= d) active.erase({pts[left].y, pts[left].x}), left++;
        auto lo = active.lower_bound({pts[i].y - d, -8e18});
        auto hi = active.upper_bound({pts[i].y + d, 8e18});
        for (auto it = lo; it != hi; ++it) {
            ll dy = pts[i].y - it->fi, dx = pts[i].x - it->se;
            best = min(best, dx * dx + dy * dy);
        }
        active.insert({pts[i].y, pts[i].x});
    }
    return best;
}
/**
* Uso: Event1D / Event2D para ordenar eventos de sweep line por coordenada x.
* Eventos para sweep line en 1D y 2D; type +1 entrada/inicio, -1 salida/fin.
* Complejidad: O(1) por evento, O(K log K) al ordenarlos con K eventos.
*/
struct Event1D {
    ll x;
    int type, id;
    bool operator<(const Event1D& o) const { return x != o.x ? x < o.x : type > o.type; }
};
struct Event2D {
    ll x, y1, y2;
    int type; // +1 borde izq, -1 borde der
    bool operator<(const Event2D& o) const { return x != o.x ? x < o.x : type > o.type; }
};
/**
* Uso: polar_sort(puntos, centro);
* Ordena puntos por angulo polar alrededor de un centro, en sentido antihorario desde el eje +x.
* Complejidad: O(N log N).

```

```

*/
template<typename T> void polar_sort(vector<Point<T>>&
    pts, Point<T> center = {}) {
    auto half = [](Point<T> p) { return p.y > 0 || (p.y ==
        0 && p.x > 0) ? 1 : -1; };
    sort(all(pts), [&](Point<T> a, Point<T> b) {
        auto va = a - center, vb = b - center;
        int ha = half(va), hb = half(vb);
        if (ha != hb) return ha > hb;
        T cr = cross(va, vb);
        if (cr != 0) return cr > 0;
        return norm2(va) < norm2(vb);
    });
}

/**
 * Uso: dist_to_line(p, a, b), dist_to_segment(p, a, b),
 *       line_intersect(a, b, c, d);
 * Distancias de un punto a una linea o segmento e
 * interseccion exacta de dos lineas infinitas no
 * paralelas.
 * Complejidad: O(1) por operacion.
 */
template<typename T> ld dist_to_line(Point<T> p, Point<T>
    > a, Point<T> b) {
    return abs((ld)cross(p, a, b)) / norm(a - b);
}

template<typename T> ld dist_to_segment(Point<T> p,
    Point<T> a, Point<T> b) {
    if (a == b) return norm(p - a);
    if (dot(p - a, b - a) <= 0) return norm(p - a);
    if (dot(p - b, a - b) <= 0) return norm(p - b);
    return dist_to_line(p, a, b);
}

// Interseccion exacta de lineas infinitas AB y CD (no
// paralelas)
Point<ld> line_intersect(Point<ld> a, Point<ld> b, Point
    <ld> c, Point<ld> d) {
    ld cr1 = cross(b - a, c - a), cr2 = cross(b - a, d - a
    );
    return c + (d - c) * (cr1 / (cr1 - cr2));
}

/**
 * Uso: rotating_calipers(hull);
 * Diametro al cuadrado de un poligono convexo mediante
 * calipers rotativos.
 * Complejidad: O(N).
 */
template<typename T> T rotating_calipers(const vector<
    Point<T>>& poly) {
    int n = poly.size();
    if (n < 2) return 0;
    if (n == 2) return norm2(poly[0] - poly[1]);
    T best = 0;
    int j = 1;
    for (int i = 0; i < n; i++) {
        int ni = (i + 1) % n;
        while (abs(cross(poly[j], poly[i], poly[ni])) < abs(
            cross(poly[(j + 1) % n], poly[i], poly[ni]))) j = (
            j + 1) % n;
        best = max({best, norm2(poly[i] - poly[j]), norm2(
            poly[ni] - poly[j])});
    }
}

```

```

}
return best;
}

/**
 * Uso: Circle c(centro, radio); circle_line_intersect(a
 *       , b, c); circle_circle_intersect(c1, c2);
 * Circulo e intersecciones con lineas y otros circulos;
 * devuelven 0, 1 o 2 puntos.
 * Complejidad: O(1) por operacion.
 */
struct Circle {
    Point<ld> c;
    ld r;
    Circle(Point<ld> c = {}, ld r = 0) : c(c), r(r) {}
};

vector<Point<ld>> circle_line_intersect(Point<ld> a,
    Point<ld> b, Circle circ) {
    auto dir = b - a;
    ld len2 = norm2(dir);
    if (cmp(len2) == 0) return {};
    ld t = dot(circ.c - a, dir) / len2;
    auto proj = a + dir * t;
    ld d2 = norm2(circ.c - proj);
    if (cmp(d2, circ.r * circ.r) > 0) return {};
    if (cmp(d2, circ.r * circ.r) == 0) return {proj};
    ld off = sqrt1(max((ld)0, circ.r * circ.r - d2) / len2
    );
    return {proj - dir * off, proj + dir * off};
}

vector<Point<ld>> circle_circle_intersect(Circle c1,
    Circle c2) {
    auto dir = c2.c - c1.c;
    ld d = norm(dir);
    if (cmp(d, c1.r + c2.r) > 0 || cmp(d, abs(c1.r - c2.r)
    ) < 0 || cmp(d) == 0) return {};
    ld a = (c1.r * c1.r - c2.r * c2.r + d * d) / (2 * d);
    ld h = sqrt1(max((ld)0, c1.r * c1.r - a * a));
    auto p2 = c1.c + dir * (a / d);
    auto perp = Point<ld>(-dir.y, dir.x) * (h / d);
    if (cmp(h) == 0) return {p2};
    return {p2 + perp, p2 - perp};
}

/**
 * Uso: vector<Halfplane> H; H.pb(Halfplane(a, b));
 * halfplane_intersection(H);
 * Interseccion de semiplanos (valida a la izquierda de
 * cada recta a → b); devuelve el poligono convexo
 * resultante.
 * Complejidad: O(N log N).
 */
struct Halfplane {
    Point<ld> p, pq; // p = punto, pq = vector direccional
    ld angle;
    Halfplane() {}
    Halfplane(Point<ld> a, Point<ld> b) : p(a), pq(b - a),
        angle(atan2l(pq.y, pq.x)) {}
    bool out(Point<ld> r) const { return cross(pq, r - p)
        < -EPS; } // r afuera (lado derecho estricto)
    bool operator<(const Halfplane& e) const {
        if (abs(angle - e.angle) > EPS) return angle < e.
            angle;
    }
}

```

```

    return cross(pq, e.p - p) > EPS;
}

Point<ld> intersect(Halfplane s, Halfplane t) {
    ld alpha = cross(t.p - s.p, t.pq) / cross(s.pq, t.pq);
    return s.p + s.pq * alpha;
}

vector<Point<ld>> halfplane_intersection(vector<
    Halfplane>& H) {
    sort(all(H));
    int n = H.size(), k = 0;
    vector<Halfplane> U(n);
    for (int i = 0; i < n; i++) {
        if (i > 0 && abs(H[i].angle - H[i - 1].angle) < EPS)
            continue;
        U[k++] = H[i];
    }
    H = U; H.resize(k); n = k;
    deque<Halfplane> dq;
    dq.push_back(H[0]);
    if (n > 1) dq.push_back(H[1]);
    for (int i = 2; i < n; i++) {
        while (dq.size() >= 2 && H[i].out(intersect(dq.back
            (), dq[dq.size() - 2]))) dq.pop_back();
        while (dq.size() >= 2 && H[i].out(intersect(dq.front
            (), dq[1]))) dq.pop_front();
        dq.push_back(H[i]);
    }
    while (dq.size() >= 3 && dq.front().out(intersect(dq.
        back(), dq[dq.size() - 2]))) dq.pop_back();
    while (dq.size() >= 3 && dq.back().out(intersect(dq.
        front(), dq[1]))) dq.pop_front();
    vector<Point<ld>> poly;
    if (dq.size() < 3) return poly;
    for (int i = 0; i < dq.size(); i++) poly.pb(intersect(
        dq[i], dq[(i + 1) % dq.size()]));
    vector<Point<ld>> clean; // quita vertices duplicados
    // y colineales
    for (int i = 0; i < poly.size(); i++) {
        auto cur = poly[i], nxt = poly[(i + 1) % poly.size()
        ];
        if (hypot(cur.x - nxt.x, cur.y - nxt.y) < EPS)
            continue;
        if (!clean.empty()) {
            auto prv = clean.back();
            if (abs(cross(cur - prv, nxt - cur)) < EPS)
                continue;
        }
        clean.pb(cur);
    }
    return clean;
}

/**
 * Uso: Point3D p(x, y, z); dist_point_plane(p, a, n);
 *       dist_point_line(p, a, b); line_plane_intersect(L1,
 *       L2, P, N);
 * Punto 3D con producto punto y cruz, distancias punto-
 * plano y punto-linea e interseccion linea-plano.
 * Complejidad: O(1) por operacion.
 */
struct Point3D {
    double x, y, z;
}

```

```

Point3D(double x = 0, double y = 0, double z = 0) : x(
    x), y(y), z(z) {}
Point3D operator+(const Point3D& p) const { return {x
    + p.x, y + p.y, z + p.z}; }
Point3D operator-(const Point3D& p) const { return {x
    - p.x, y - p.y, z - p.z}; }
Point3D operator*(double c) const { return {x * c, y *
    c, z * c}; }
double dot(const Point3D& p) const { return x * p.x +
    y * p.y + z * p.z; }
Point3D cross(const Point3D& p) const { return {y * p.
    z - z * p.y, z * p.x - x * p.z, x * p.y - y * p.x};
}
double length() const { return sqrt(dot(*this)); }
};
double dist_point_plane(Point3D p, Point3D a, Point3D n)
{ return abs((p - a).dot(n)) / n.length(); }
double dist_point_line(Point3D p, Point3D a, Point3D b)
{ return (p - a).cross(b - a).length() / (b - a).
    length(); }
Point3D line_plane_intersect(Point3D L1, Point3D L2,
    Point3D P, Point3D N) {
    auto u = L2 - L1;
    double d = N.dot(u);
    if (abs(d) > EPS) return L1 + u * ((P - L1).dot(N) / d
    );
    return {1e9, 1e9, 1e9}; // paralelos
}

```

Bit Operations

```

/**
 * Uso: check_bit(m, i), set_bit(m, i), clear_bit(m, i),
 * toggle_bit(m, i), lsb(m). i 0-based.
 * Manipulacion basica de bits de una mascara: consultar
 * activar, limpiar, alternar y aislar el bit menos
 * significativo.
 * Complejidad: O(1) por operacion.
 */
inline bool check_bit(ll mask, int i) { return (mask >>
    i) & 1; }
inline ll set_bit(ll mask, int i) { return mask | (1LL
    << i); }
inline ll clear_bit(ll mask, int i) { return mask & ~(1
    LL << i); }
inline ll toggle_bit(ll mask, int i) { return mask ^ (1
    LL << i); }
inline ll lsb(ll mask) { return mask & -mask; }
/**
 * Uso: count_bits(x), count_bits_ll(x), highest_bit(x)
 * = bit mas alto (-1 si 0), ctz(x) = ceros a la
 * derecha.
 * Conteo de bits activos, posicion del bit mas alto y
 * numero de ceros finales con instrucciones nativas.
 * Complejidad: O(1) por operacion.
 */
inline int count_bits(int mask) { return
    __builtin_popcount(mask); }
inline int count_bits_ll(ll mask) { return
    __builtin_popcountll(mask); }
inline int highest_bit(ll mask) { return mask == 0 ? -1
    : 63 - __builtin_clzll(mask); }

```

```

inline int ctz(ll mask) { return mask == 0 ? 64 :
    __builtin_ctzll(mask); }
/**
 * Uso: iterate_submasks(m);
 * Itera todas las sub-mascaras de una mascara dada en
 * orden decreciente (sin incluir la vacia).
 * Complejidad:  $O(3^N)$  en total para una mascara de N
 * bits.
 */
inline void iterate_submasks(ll mask) {
    for (ll sub = mask; sub; sub = (sub - 1) & mask) { /*
        procesar sub */ }
}
/**
 * Uso: auto masks = gossers_hack(k, n);
 * Genera todas las mascaras de tamano n con exactamente
 * k bits activos (gossers hack).
 * Complejidad:  $O(\binom{n}{k})$  mascaras generadas, O(1)
 * amortizado por mascara.
 */
inline vector<ll> gossers_hack(int k, int n) {
    vector<ll> res;
    if (!k) return {};
    ll mask = (1LL << k) - 1, limit = 1LL << n;
    while (mask < limit) {
        res.pb(mask);
        ll c = mask & -mask, r = mask + c;
        mask = (((r ^ mask) >> 2) / c) | r;
    }
    return res;
}
/**
 * Uso: XorBasis b; b.insert(x); b.can_form(k); b.
 * get_max(init); b.get_min();
 * Base lineal de xor sobre enteros de 64 bits: maximo
 * xor de un subconjunto, minimo y test de
 * representabilidad.
 * Complejidad: insercion y consultas O(60).
 */
struct XorBasis {
    static const int BITS = 60;
    ll basis[BITS] = {};
    int sz = 0;
    // Inserta; true si fue linealmente independiente
    bool insert(ll x) {
        for (int i = BITS - 1; i >= 0; i--) if (check_bit(x,
            i)) {
            if (!basis[i]) return basis[i] = x, sz++, true;
            x ^= basis[i];
        }
        return false;
    }
    bool can_form(ll k) {
        for (int i = BITS - 1; i >= 0; i--) if (check_bit(k,
            i)) {
            if (!basis[i]) return false;
            k ^= basis[i];
        }
        return true;
    }
}

```

```

}
// Maximo de (initial ^ subset)
ll get_max(ll initial = 0) {
    for (int i = BITS - 1; i >= 0; i--) if ((initial ^
        basis[i]) > initial) initial ^= basis[i];
    return initial;
}
ll get_min() {
    for (int i = 0; i < BITS; i++) if (basis[i]) return
        basis[i];
    return 0;
}
/**
 * Uso: BitTrie t; t.insert(x); t.max_xor(x); t.insert(x
    , -1) borra.
 * Trie binario para consultar el maximo xor de un valor
 * contra el conjunto insertado, con soporte de
 * borrado.
 * Complejidad: insercion y consulta O(B) con B = 30
 * bits (60 si los valores llegan a  $10^{18}$ ).
 */
struct BitTrie {
    static const int BITS = 30; // 60 si los numeros
    // llegan a  $1e18$ 
    struct Node { int next[2] = {-1, -1}; int cnt = 0; };
    vector<Node> t;
    BitTrie() { t.emplace_back(); }
    void insert(ll x, int val = 1) { // val=-1 para borrar
        int u = 0;
        for (int i = BITS - 1; i >= 0; i--) {
            int b = check_bit(x, i);
            if (t[u].next[b] == -1) t[u].next[b] = t.size(), t
                .emplace_back();
            u = t[u].next[b];
            t[u].cnt += val;
        }
    }
    ll max_xor(ll x) {
        int u = 0; ll ans = 0;
        for (int i = BITS - 1; i >= 0; i--) {
            int b = check_bit(x, i), opp = b ^ 1;
            int v = t[u].next[opp];
            if (v != -1 && t[v].cnt > 0) ans |= 1LL << i, u =
                v;
            else u = t[u].next[b];
        }
        return ans;
    }
};

```

Extra

```

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
/**
 * Uso[0]: ordered_set<T> s; s.insert(x); s.order_of_key
    (x); *s.find_by_order(k);
 * Conjunto ordenado de pbds con consultas por rango (k-
    esimo menor y posicion).
 * Complejidad: O(log N) por operacion.

```

```

*/
template <typename T>
using ordered_set = __gnu_pbds::tree<T, __gnu_pbds::
    null_type, less<T>, __gnu_pbds::rb_tree_tag,
    __gnu_pbds::tree_order_statistics_node_update>;
/**
 * Uso[0]: ordered_multiset<T> s; s.insert({x, id});
 * Conjunto ordenado de pbds que admite duplicados
 * emparejando cada valor con un id unico.
 * Complejidad: O(log N) por operacion.
 */
template <typename T>
using ordered_multiset = __gnu_pbds::tree<pair<T, int>,
    __gnu_pbds::null_type, less<pair<T, int>>,
    __gnu_pbds::rb_tree_tag, __gnu_pbds::
    tree_order_statistics_node_update>;

/**
 * Uso: unordered_map<LL, int, custom_hash> m;
 * Functor de hash anti-colision (splitmix64 con semilla
 * aleatoria) para tablas hash de claves numericas.
 * Complejidad: O(1) por operacion de hash.
 */
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM = chrono::
            steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

/**
 * Uso: MeetInTheMiddle::subset_sum(a, target);

```

```

* Resuelve subset sum exacto con la tecnica meet in the
  middle para arreglos de hasta 40 elementos.
* Complejidad:  $O(N \cdot 2^{N/2})$  tiempo,  $O(2^{N/2})$  espacio.
*/
struct MeetInTheMiddle {
    static bool subset_sum(const vll& a, ll target) {
        int n = a.size(), mid = n / 2;
        vll left, right;
        for (int i = 0; i < (1 << mid); i++) {
            ll s = 0;
            for (int j = 0; j < mid; j++) if (i >> j & 1) s +=
                a[j];
            left.pb(s);
        }
        int n2 = n - mid;
        for (int i = 0; i < (1 << n2); i++) {
            ll s = 0;
            for (int j = 0; j < n2; j++) if (i >> j & 1) s +=
                a[mid + j];
            right.pb(s);
        }
        sort(all(right));
        for (ll s : left) if (binary_search(all(right),
            target - s)) return true;
        return false;
    }
};

/**
 * Uso[0]: ZobristHash zh; auto pref = zh.build_pref(a);
 * zh.query_range(pref, l, r);
 * Hash de prefijos xor para comparar multiconjuntos de
 * rangos en O(1).
 * Complejidad: construccion O(N), consulta O(1).
 */
struct ZobristHash {
    using u64 = uint64_t;
    mt19937_64 rng;
    unordered_map<int, u64> val_hash;

```

```

ZobristHash(u64 seed = 1337) : rng(seed) { val_hash[0]
    = 0; }
u64 get_hash(int x) {
    if (!val_hash.count(x)) val_hash[x] = rng();
    return val_hash[x];
}
vector<u64> build_pref(const vi& a) { // prefijos XOR
    vector<u64> pref(a.size() + 1, 0);
    for (int i = 0; i < a.size(); i++) pref[i + 1] =
        pref[i] ^ get_hash(a[i]);
    return pref;
}
u64 query_range(const vector<u64>& pref, int l, int r)
    { return pref[r + 1] ^ pref[l]; }
};

/**
 * Uso[1]: BitsetReachability<MAXN> br; br.build(n,
 * aristas); br.can_reach(u, v);
 * Cierre transitivo (alcanzabilidad todos-pares) de un
 * grafo dirigido usando bitsets.
 * Complejidad: construccion  $O(N^3/64)$ , consulta O(1).
 */
template<size_t MAXN>
struct BitsetReachability {
    bitset<MAXN> reachable[MAXN];
    void build(int n, const vector<pii>& edges) {
        for (int i = 1; i <= n; i++) reachable[i].reset(),
            reachable[i][i] = 1;
        for (auto [u, v] : edges) reachable[u][v] = 1;
        for (int k = 1; k <= n; k++)
            for (int i = 1; i <= n; i++)
                if (reachable[i][k]) reachable[i] |= reachable[k];
    }
    bool can_reach(int u, int v) { return reachable[u][v]; }
};

```